MILITARY SPECIFICATION
SECOND ORDER ATTITUDE REFERENCE SYSTEM (SOARS)
AN/ASN-116

This specification has been approved
by the Naval Air Systems Command,
Department of the Navy

1. SCOPE

1.1 Scope - The equipment conforming with this specification shall provide a compensated all attitude reference system for pitch, roll, and heading angular displacements utilizing dynamic second order erection with velocity reference and all the performance requirements described herein.

1.2 Classification - The equipment of this specification shall consist of the following components:

Type Designation	Applicable Paragraph
System SOA2S, AN/ASN-116	3.5
Gyroscope, Displacement, CN-1434/ASN-116	3.5.2
Amplifler, Electronic Control, AM-6674/ASN-116	3.5.3
Control, Compess C-9550/ASN-116	3.5.4
Compensator, Electronic, Magnetic Compass CN-1492/ASN-110	3.5.5

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.7.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein. When there is a conflict between referenced documents and this specification, this specification will take precedence.

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SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Airborne General Specification for
MIL-T-5422	Testing Environmental, Airborne, Electronic and Associated Equipment
MIL-C-6781	Control Panel, Aircraft Equipment, Rack on Console Mounted
MIL-P-7788	Plate, Plastic, Lighting
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Eletrical Equipment and Associated Repair. Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Reproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-23103	Thermal Performance Evaluation, Airborne, Electronic
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment
MIL-L-25467	Lighting, lnstrument, Integral , General Specification for
MIL-L-27160	Lighting, instrument, Integral, White, General Specification for
MIL-G-38134	Detector, Azimuth, Magnetic DSU-4A/A
MIL-C-38999	Miniature, Circular, Environment Resisting, Series I
MIL-G-0081559B	Gyroscope, Assemblies, Attitude and Directional Reference Instruments for Aircraft; Preservation, Packaging and packing
MIL-B-81744	Barrier Coating Solution and Solvent
MIL-B-81793	Bearing, Ball, Precision for Instruments and Rotating Components

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Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-461	Electromagnetic, Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic, Interference Characteristics, Measurement of
MIL-STD-470	Maintainability Program Requirements
MIL-STD-471	Maintainability Verification/Demonstration Evaluation
NIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment, Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging of
MIL-STD-810	Environmental Test Methods
MIL-STD-1304	Reliability Reports
MIL-STD-1334	Process for Barrier Coating of Anti-Friction Bearings
MIL-STD-2063(AS)	Reliability Development Tests
MIL-STD-2074(AS)	Failure Classification for Reliability Testing
MIL-STD-2076(AS)	Unit Under Test Compatibility with Automatic Test Equipment, General Requirements for
MIL-STD-2077(AS)	Test Program Sets, General Requirements for
MS17322	Meter, Time Totalizing, Miniature Digital, 115V, 400 Hz
MS27651	Meter, Time Totalizing Hermetically Sealed, 115V, 400 Hz
	Numerals and Letters, Aircraft Instrument Dial, Standard Form of

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2.1.1 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(a) Microelectronic Technology shall be considered and microelectronic items shall conform to requirements specified herein.

(b) All parts and materials requirements shall conform to Specification MIL-E-5400, Revision L.

(c) Nonrepairable subassemblies shall be used in accordance with Specification AR-10 and as outlined in Specification MIL-E-5400.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials not previously government approved (including electron tubes, transistors, diodes, and microelectronic devices) shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Microelectronic Modular Assemblies - Conformal coatings, encapsulents embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removed without damage to the assembly.

3.2.3 Modules - The electronic portions of the equipment shall be functionally modularized in accordance with AR-10.

3.3 Design and Construction - The equipment shall conform with all applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall be 29.5 pounds.

3.3.1.1 Form Factor - The total volume of Second Order Attitude Reference System, excluding cables and mounts, shall not exceed 825 cubic inches. The external surfaces except as specified herein shall be painted lusterless gray, color number 36231, in accordance with FED-STD-595.

3.3.2 Reliability - The contractor shall conduct a reliability test program in accordance with MIL-STD-785.

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3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 2500 hours without the necessity for readjustment of any controls that are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failures (MTBF) - The system, including any Built-in-Test provisions, shall have 1250 hours of mean time between failures (MTBF) when tested and accepted as outlined under the requirements of 4.4.3. The MTBF of the components shall be: Displacement Gyroscope 3000 hours, Electronic Control Amplifier 2500 hours, Controller 30,000 hours and Electronic Compass Compensator 30,000 hours.

3.3.2.4 Time Totalizing Meter - The following units shall contain time totalizing meters in accordance with Specification Mil-M-7793.

Unit	Type of Meter
Displacement Gyroscope Assembly	MS27651
Electronic Control Amplifier	MS27651 or MS17322

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-N-5088. The external wiring shall be unshielded, except that a numeber of the individual wires shall be shielded when necessary to meet interference control requirements. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment specified herein. The interconnection cabling shall be in accordance with Figure 7.

3.3.3.3 Electrical Connectors - External connectors on the SOARS components shall be in conformance with MIL-C-38999 Series 1. All component connectors shall be protected by connector caps during shipment.

3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of specification MIL-C-6781. The configuration of all' panels must be approved by the procuring activity.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3.3.6 Electromagnetic Interference - The emission and susceptibility of equipment conforming herein to electromagnetic interference shall meet CEO1, CE03, CE04, CSO1, CS02, CS06, REO1, RE02, RSO1, RS02, RS03, Transient Spike Generation,and Relay Transient Susceptibility requirements of MIL-STD-461.

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3.3.7 Provisions for Maintainability - The Maintainability Program, built-in test equipment features, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in Specification AR-10. The following maintainability requirements shall be considered in the set and component design.

(1) Minimization of the complexity of maintenance tasks (for example: calibration, adjustments, inspection, etc) by maximum use of a design which includes optimum interchangeability and use of standardized equipment or commercial items.

(2) Rapid and positive recognition of equipment malfunction or marginal performance.

(3) Rapid and positive identification of the replaceable defective part, assembly or component.

(4) Minimum numbers and types of tools and test equipment needed to perform maintenance.

(5) Optimum accessibility in all systems, equipments, and components requiring maintenance, inspections, removal or replacement.

3.3.7.1 Maintenance Procedures - When specified by the contract, the contractor shall establish a complete maintenance procedure which shall be submitted to the procuring activity for approval at the time specified in the contract. Maintainability provisions shall conform to the requirements of MIL-STD-470.

3.3.7.1.1 Maintainability Prediction Reports - When specified by the contract, the contractor shall prepare maintainability prediction reports in accordance with AR-10.

3.3.7.1.2 Compatibility with VAST - The equipment shall be compatible with the Automatic Test Equipment (ATE) and shall meet the requirements of MIL-STD-2076(AS). When required Test Program Sets shall be furnished in accordance with MIL-STD-2077(AS). If Test Program Sets exist for the equipment, and changes to the equipment are made that will effect the fault diagnosis procedure, changes to the existing Test Program Sets shall be prepared as part of the equipment changes in accordance with MIL-STD-2077(AS).

3.3.8 Nomenclature, Nameplates and Identification Marking - Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making bench tests.

Temperature	Room ambient (25°C plus or minus 10°C)
Magnetic Field	.18 oersted horizontal .54 oersted vertical

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Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 95% relative humidity
Input Power Voltage	115/200 plus or minus 1.5 V ac, 400 Hz, 3 phase
	Panel lighting voltage, 5 ± .1 V ac, 400 Hz

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment.

3.3. 10.1 Acoustical Noise - The equipment shall operate satisfactorily after being subjected to an acoustical noise sound pressure level of 140 dB for a period of 30 minutes.

3.3.11 Reaction or Warm-Up Time - The time required for the system to operate shall be kept to a minimum and shall not exceed 1.0 minute under standard conditions and 1.5 minutes at extreme service conditions.

3.3.12 Input Electrical Power -

3.3.12.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amounts.

Alternating Current Power (3-Phase, 4-Wire), 115/200 volts, Category B, 300 volt-amperes maximum, 200 volt-amperes after 90 seconds.

3.3.12.2 Lighting Power - Input power for lighting shall require not more than 1.5 amperes at 5 volts. (7.5 volt amperes)

3.3.12.3 Degarded Performance. Degraded performance will be permitted for voltage transients only as specified in MiL-STD-704 (Category B). System operation shall return to normal with no resulting damage to the equipment.

3.3.13 Cooling -

3.3.13.1 Thermal Conditions - The system shall operate satisfactorily under the thermal conditions specified in MIL-E-5400 Class 2.

3.3.13.2 Operating Temperature Check Point - The Electronic Control Amplifier (ECA) shall have a point accessible on the outer surface of the equipment for monitoring the temperature(s) of the most critical part(s) inside. The operating temperature(s) of the part(s) selected for the check point shall be determined from the evaluation by MIL-T-23103. This check point shall be clearly marked and shall be incorporated into equipment delivered not later than 45 days after the procuring activity's approval of the cooling design data.

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3.3.14 Gyroscope Bearings -

3.3.14.1 Rotor Bearing Lubrication - The rotor bearing shall be lubricated with Mobil RL-/43A oil conforming to MIL-B-81793.

3.3.14.2 Bearing Lubrication - The lubricating oil used in gimbal bearings shall be Mobil-7 4 3 A conforming to MIL-B-81793.

3.3.14.3 Barrier Film Bearing Coating - All Rotorace motor bearings shall be processed in accordance with MIL-STD-1334. The coating material shall be in accordance with MIL-B-81744. All Rotorace motor bearings, including chamfered races or corners, and including interfacing parts where applicable, shall be barrier coated.

3.3.14.4 Ball Bearings - Ball bearings shall be ABEC-5 grade or better. Teflon separators shall be used on gimbal bearings where the bearing design permits. Spin rotor bearings shall have porous separators impregnated with Mobil RL-743A per MIL-B-81793 and shall be ABEC-7 or better.

3.3.14.5 Ball Bearing Pre-Loading - Means shall be provided for preloading an alignment without causing eccentric loading unless otherwise authorized.

3.4 Performance - Unless otherwise specified in the approved test plans, values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The Second Order Attitude Reference System (a flight safety reference system) shall be an all attitude source of roll, pitch and heading angular displacement signals for other airborne installed equipments, Including flight control and reference, navigation, weapons deliveries and direct presentation or displays. Roll information shall be continuous throughout 360 degrees and valid for all aircraft angles of pitch and heading. Pitch information shall be continuous throughout plus or minus 90 degrees and valid for all angles of roll or heading. Heading information shall be continuous throughout 360 degrees and valid for all attitudes of pitch and roll. There are no stops or controlled precession points in the attitude or heading axes of the SOARS. The operativg ranges for SOARS shall be as follows:

Parameter	Mission Condition
Latitude Range (degrees)	S90 to N90
Aircraft Reference Velocity (knots)	0 to 1000
Aircraft Roll Attitude (degrees)	-180 to +180
Aircraft Elevation (degrees)	-90 to +90
Aircraft Heading (degrees)	0 to 360 (continuous)

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Parameter	Mission Condition
Aircraft Yaw Angular Tate (degrees/second)	o to 200
Aircraft Roll Angular Rate (degrees/second)	0 to 600
Aircraft pitch Angular Rate (degrees/second)	0 to 100
Aircraft Pitch Angular Rate through Zenith (degrees/second)	0 to 50
Aircraft Roll Angular Acceleration (degrees/second squared)	0 to 1500
Aircraft Pitch and Yaw Angular Acceleration (degrees/second squared)	0 to 200
Linear Acceleration Range (feet/second squared)	0 to 320 (any direction)

Vertical erection shall be provided by the compensated second order erection mode using internal accelerometers and an external velocity reference. In the event of loss of all velocity information, the system shall revert to a compensated first order mode of vertical erection. Heading data shall be available in all modes. In the DG mode, heading shall be derived from the directional gyro which shall be compensated for earth's rate and mean drift; in the SLAVED mode, the directional gyro shall be slaved to the earth's magnetic field as sensed by the magnetic azimuth detector or transmitter. In the EMERGENCY mode the output of the magnetic azimuth detector shall be provided with electronic damping. The SOARS shall provide a turn rate signal for display equipment. The SLAVED mode directional performance shall be improved by providing adjustable slaving cutoff values for turns and longitudinal accelerations. Normally, the SOARS shall operate in a velocity damped condition utilizing valid true airspeed as the velocity reference. In the event true airspeed information is determined invalid by either the invalid discrete signal or the data processor software techniques, the SOARS shall revert to compensated first order mode within 30 ± 5 seconds. After velocity information has been redetermined valid, the system shall return to the Second Order velocity damped condition within 2 ± 2 seconds. The backup first order mode shall be configured to include earth's rate input to the vertical gyro. Cutout shall be mechanized as follows:

Roll Erection Cutout	Turn rate greater than or equal to 8.0 degree/minute or lateral acceleration greater than .07g
Pitch Erection Cutout	Lateral acceleration greater than .35g or longitudinal acceleration greater than .07g
Slaving Cutout	Turn rate greater than or equal to 8.0 degree/minute or lateral acceleration greater than .07g or longitudinal acceleration greater than .07g or pitch greater than 25 degrees or roll greater than 25 degrees

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The heading turn rate for magnetic slaving cutoff shall be the same regardless of the vertical mode. Cutoff shall be commanded within 30 seconds after the start of an 3 degree per minute turn, and erection shall be restored within 20 seconds after termination of the turn rate. Cutoff shall not be actuated by a 3 degree peak to peak oscillation with an 8 second period nor by a 1.5 degree peak to peak oscillation with a period of one second. Turn rate cutoff shall be disabled in the EMERG mode.

All system operational computations shall be done digitally. All system computations shall be primarily digital with analog output, as specified, for the aircraft equipments, controls, displays and presentations. Provision shall be included in the digital processor for reducing digital pulse train and sense information for reference velocity utilization. Acceleration in either direction along the vertical axis greater than .07g shall be used to cutoff the respective erection as specified herein for the compensated first order mode. A timer shall be actuated that disables the cutoff command if the condition exists for more than 360 seconds. After the acceleration has dropped below the cutoff threshold for 15 seconds, the timer shall be reset and the cutoff disable shall be removed.

3.4.2 Accuracies - The systm shall be capable of providing output and requirements with the accuracies indicated below:

VERTICAL

Output	Static Accuracy	Dynamic Accuracy**
Velocity Damped		
Pitch and Roll	.20 degree max	.25 degree rms
Verticality		.70 degree max
First Order		
Pitch and Roll	.20 degree max	1.5 degrees
Verticality		3.0 degree max

HEADING

Mode	Static Accuracy	Dynamic Accuracy
DG	.5 degree max	(1.0 degree/hr max)
SLAVED	.2 degree max	.4 degree rms
EMERGENCY	1.0 degree max	1.0 degree max***

GYROSCOPICS

Directional Gyro	Random	.25 deg/hr rms
	Day to Day Bias	±1.0 deg/hr max
Vertical Gyro		6.0 deg/hr max each axis
Accelerometers	Dynamic	±2.0g
	Linearity	Figure 1

***prevailing wind not greater than 100 knots
***with respect to the transmitter output

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3.4.3 Inputs - The system shall require the following inputs from other aircraft equipments:

3.4.3.1 Velocity Reference - The SOARS shall require a velocity reference, true air speed, in analog form for velocity damped Second Order operation.

3.4.3.2 True Air Speed Velocity Input - The air data system input when provided shall be true air speed along aircraft heading in analog form. The data processor in the Electronic Control Amplifier shall process all the true airspeed parameters for use in the velocity damped condition.

The true airspeed signal shall have the following characteristics:

Signal Format - Analog

Source in ADC - 2K ohms linear potentiometer

Output Range - 200 to 750 knots

Output Scaling - 1.589% Vref/10 kt

Scaling Datum- 1.9% Vref - 200 kt

The potentiometer should be energized by a voltage (Vref), from the Electronic Control Amplifier.

3.4.3.3 True Air Speed Velocity Invalid Discrete - A separate signal shall be provided to indicate the invalidity of the true air speed data signal to the system.

The true airspeed valid discrete shall be provided with the following characteristics:

Signal Format - Discrete

Source in ADC - Single Pole Switch (.5 amp rated)

Switch open circuit - TAS invalid

Switch short circuit - TAS valid

3.4.3.4 Magnetic Heading- Magnetic heading information shall be supplied to the system by a Magnetic Azimuth Detector as described in Paragraph 6.7. Excitation for the compass transmitter shall be supplied by the SOARS as required by MIL-D-38134.

3.4.4 Outputs - The equipment shall provide the following outputs:

3.4.4.1 Analog Heading Output - Four synchro outputs shall be provided by the equipment. Two external 115 V ac, 400 Hz excitations shall be required. One excitation shall supply three of the outputs and the second excitation shall supply the remaining heading outputs. One of the three commonly excited heading outputs shall have a solid state output with the Z-leg grounded internal to the Electronic Control Amplifier. The other two of the three commonly excited outputs shall be electrically isolated from all other system outputs. The

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fourth output, excited from the second source, shall be electrically isolated from all other outputs.

3.4.4.2 Analog Pitch and Roll Outputs - One electrically isolated, commonly excited equivalent synchro output shall be provided for pitch and roll signals.

3.4.4.3 Turn Rate Output - The equipment shall provide a derived rate of heading output which shall be compatible with current rate of turn indicators.

3.4.4.4 Flight Control Decouple - The equipment shall provide a flight rate control decouple signal which shall indicate a disengage condition when the system is operated in the EMERG mode, the SET HDG control is activated in the DG mode, or when the FAST SYNC function is utilized in the SLAVED mode.

3.4.4.5 System Status Signals - The equipment shall provide status signals which indicate the validity of the attitude information and the BIT status.

3.4.4.5.1 Invalid Discrete - The invalid discrete shall indicate an invalid condition if the system is in the initialization sequence, if the fast erect function has been activated, or if a failure is detected by BIT.

3.4.4.5.2 Malfunction Discrete - The malfunction discrete shall indicate a failed condition for any BIT detected failure.

3.4.5 Modes of Operation - Mode of operation for the system shall be manually selected by positioning the Mode Selector switch on the Compass Controller to SLAVED mode, DG mode, or EMERGENCY mode.

3.4.5.1 SLAVED Mode - When the system is operated in the SLAVED mode, the heading outputs shall provide earth rate compensated, gyroscopically stabilized magnetic headings.

3.4.5.2 DG Mode - When operated in the DG mode, the equipment shall provide earth rate corrected Directional Gyro heading.

3.4.5.3 EMERGENCY Mode - When operated in the EMERGENCY mode, the heading outputs shall be electronically damped and unstabilized magnetic heading.

3.4.6 Acceleration - Internal accelerometer sensors on the gyroscopic assembly, along with the associated circuitry, shall provide heading slaving cutoff for acceleration in excess of .07g in either longitudinal or lateral axis. Heading slaving shall be cut off for an aircraft attitude outside the envelope of plus or minus 25 degrees in either pitch or roll.

3.4.7 Erection Control -

3.4.7.1 Compensated First Order Operation - The system shall be in a compensated first order erection mode during the following conditions:

(1) Initial vertical gyro erection sequence.

(2) Invalid discrete signal or data processor software indicating the velocity reference is unreliable or invalid.

(3) Emergency mode of heading operation.

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3.4.7.1.1 Reduction in Erection Rate - Acceleration in either axis greater than plus or minus .07g shall be used to initiate erection cutoff and timing per 3.4.1.

3.4.7.1.2 Manual Fast Erection - A momentary pushbutton shall be provided that when depressed will result in the pitch and roll erection system reverting to the fast erection rate.

3.4.7.2 Second Order Operation - The system shall be in a Velocity Damped Second Order erection mode during the following conditions:

(1). The vertical gyro initial erection sequence has been completed.

(2) Velocity reference input to the system is determined valid.

(3) The system is in the SLAVED or DG mode.

3.4.7.2.1 Lateral Acceleration - Roll erection shall be cut off for lateral acceleration in excess of 1.89s. Vertical drift in the roll axis due to earth rate and mean gyro drift shall be compensated when in roll erection cutoff.

3.4.7.2.2 Longitudinal Acceleration - Pitch erection shall be cut off for longitudinal acceleration in excess of 1.8gs. Vertical drift due to earth rate, earth profile, and VG pitch drift shall be-compensated when in pitch erection cutout.

3.4.8 Built-In-Test Equipment Capabilities (BITE) -

3.4.8.1 Fault Detection - Provisions shall be made for the detection of equipment malfunctions. The built-in-test equipment shall be capable of detecting a minimum of 85 percent of all equipment malfunctions.

3.4.8.2 Fault Isolation - The built-in-test equipment shall provide a minimum of 85 percent probability of isolating a detected malfunction to the faulty weapons replaceable assembly.

3.4.9 Compensation - Heading compensations shall be provided for index, one cycle and two cycle errors. The heading compensations shall be operative in the SLAVED and EMERGENCY modes of operation.

3.4.9.1 Index Compensation - Adjustments shall be provided for all index compensation. The adjustments shall remain where set.

3.4.9.2 Single Cycle Compensation - Adjustments shall provide for single cycle, EW and NS compensations. The single cycle compensation shall be automatically self-compensating with variations of the magnetic field intensity.

3.4.9.3 Two Cycle Compensation - The two cycle compensation shall be solid state and shall require two adjustments to compensate. One adjustment shall act on cardinal heading errors and one adjustment shall act on intercardinal heading errors.

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3.4.9.4 Compensation Disabling Switch - A disabling switch shall be provided to remove compensation. Means shall be provided so that compensation shall always be available when the equipment is operating.

3.4.9.5 Fast Swing Switch - A switch shall be provided to reduce the heading coupler time constant to permit accomplishment of compass calibration.

3.4.10 Pitch and Roll Maneuvers - The equipment shall be capable of operating within the specified limits during a 360 degree continuous loop or roll maneuver at the following rates:

Pitch Rate	O to 50 degrees per second inclusive
Roll Rate	O to 600 degrees per second inclusive

3.4.10.1 Profile Maneuver - The SOARS shall perform during the flight profiles specified in Figures 4, 5, and 6. During and upon completion of these maneuvers the performance shall be within the accuracies specified in 3.4.2.

3.4.11 Synchronization -

3.4.11.1 Automatic Synchronization - Upon completion of the gyro erection initialization, the system when operated in the SLAVED mode, shall-automatically synchronize the heading outputs.

3.4.11.2 Manual Synchronization - When operated in the SLAVED mode, depression of the SLT HEADING control shall cause the system to automatically synchronize the heading outputs.

3.4.11.3 Mode Switching Synchronization - When the Compass Control Mode Select switch is changed from DG mode to SLAVED mode, or DG mode to EMERGENCY mode, the system shall automatically synchronize the heading outputs.

3.4.12 Set Heading Capability - When operated in the DG mode, the SET HEADING control shall provide a continuously variable drive rate from 0 to 30 degrees per second in either direction when pushed and turned.

3.4.13 Mode Switching Heading Alignment Retention - When the SOARS system is switched from the SLAVED mode to DG mode, the heading outputs shall not change more than .1 degree.

3.5 DETAILED DESIGN REQUIREMENTS

3.5.0.1 The Second Order Attitude Reference System shall provide outputs for use by airborne flight control, bombing and navigation equipments as detailed herein. The equipment shall normally provide a dynamically compensated Second Order vertical erection, functionally utilizing an externally derived velocity reference damped with internal acceleration inputs. Should the external velocity reference fail, or be unavailable for an extended period of time, the vertical erection shall revert to a compensated in-flight first order vertical erection mode including earth rate. These effects shall be continuously applied. Second Order erection shall commence immediately when the external velocity reference signal becomes valid. The Second Order of vertical erection equipments shell be designed to assure maximum efficiency in the use of plug-in subassemblies, maximum reliability and minimum maintenance time.

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3.5.0.2 The Displacement Gyroscope shall provide angular output in pitch, roll, and azimuth. The gyroscopic reference shall consist of either single-degree-of-freedom or two-degree-of-freedom gyros and accelerometers mounted within the selected gimbal arrangement. The arrangement shall permit all-altitude operation at all times without any restrictions. No error exceeding 1.0 degree and lasting more than 5 seconds shall be developed as a result of flying in the vertical climb and/or dive positions. The accelerometers shall be linear over the total range of plus or minus 2gs. The hermetically sealed gyroscopes and electronics shall be so constructed that they shall be removable from the assembly and replaceable without use of special tools or skill.

3.5.0.3 The Electronic Control Amplifier shall supply the following functions:

(1) First and Second Order vertical erection control and functionally associated reference computations.

(2) Roll, pitch, rate of turn, and magnetic heading system outputs and circuitry.

(3) Required circuits for processing all incoming and outgoing signals required for operation and application of the system.

(4) AC/DC power supply, all vertical and directional gyro erection and control circuitry, all system interface circuits, mode control, tests, and digital logic.

(5) Self-cooling features.

(6) System functions not assigned to other components.

The amplifier cards shall be modular plug-in type. The amplifier circuit cards shall be designed to assure maximum efficiency in the use of plug-in subassemblies, with maximum reliability and minimum maintenance time.

3.5.0.4 The SOARS Controller shall provide as a minimum the following system control functions and indications with all respective reference functions grouped in the related areas:

(1) Mode Selector

(2) Fast Synchronization Command

(3) Set Heading Control

(4) Synchronization Indicator

(5) Latitude Readout and Manual Control

(6) Malfunction Naming Lamp

(7) Fast Erection Command

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(8) Magnetic Variation Input and Control

(9) Hemisphere Selection Switch

3.5.0.5 The Compass Compensator shall be a modular unit which shall be mounted in the aircraft cockpit or on an avionics rack in a manner to permit visibility and quick removal, The unit shall be of minimum size and weight. It shall be completely interchangeable and shall not require any adjustment or readjustment when one or more of the other system components are removed and replaced in an aircraft installation.

3.5.1 System Parameters - The Second Order Attitude Reference System shall be provided with the means of complying with the requirements stated herein.

3.5.1.1 Erection Cycles -

3.5.1. 1.1 Normal Operation -

3.5.1.1.1.1 Initial Erection - Fast erection in pitch and roll shall be automatically sequenced following application of power as specified herein. The vertical gyro spin axis shall assume a position within 1.0 degree of local vertical within 1 minute under standard conditions and 1.5 minutes under extreme service conditions after power is applied. The gyroscope rotors shall have been at a standstill with spin axes in any random attitude at the time of power application. Upon completion of initial erection, the system shall be operating as specified in 3.4.1. When the erection control senses a null of less than 1.0 degree after the initialiation sequence, normal erection shall begin automatically.

3.5.1.1.1.2 Final Erection - After completion of the initial or fast cycles, application of low erection voltage shall cause erection of the gyro to a position within .15 degree rms of vertical in 1.5 minutes or less at standard conditions and within 2 minutes at extreme service conditions.

3.5.1.1.2 Erection Rates -

3.5.1.1.2.1 Normal Erection - The normal erection rate about the pitch and roll axes shall be in accordance with Figure 2.

3.5.1.1.2.2 Fast Erection -The fast erection rate about the pitch and roll axes shall be a minimum of 15 degrees per minute when automatically or manually initiated. The Controller shall provide means for external manual operation of the fast erection circuits. Activation of the Controller fast erection switch shall control the fast erection. Roll fast erection shall be prevented if the system is in a turn rate cutoff function.

3.5.1.1.3 Erection Control - First Order

3.5.1.1.3.1 Fast Erection Command - A momentary pushbutton shall be provided on the Controller. When depressed it will result in the pitch and roll erection system reverting to the fast erection rate. If during this period the roll and pitch errors approach within .5 degree of the vertical, the erection shall immediately revert to normal erection. The amplifier monitor signal shall indicate only a temporary invalid at any time the anplifier and system are in

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fast erection. Protective devices shall be provided which prevent application when higher design or command functions are in operation or when thermal damage can occur.

3.5.1.1.3.2 Automatic Fast Erection - Fast erection shall be automatically initiated after initial power application and after loss of power as specified herein. Fast erection shall be terminated and the system shall revert to normal First Order erection when roll and pitch errors are detected to be no greater than plus or minus .5 degree and the time since power has been applied is one minute minimum.

3.5.1.1.3.3 Lateral Acceleration -An accelerometer mounted on the Displacement Gyroscope with associated circuitry shall be provided to cut off the roll erection during horizontal turn accelerations. For aircraft lateral accelerations of plus or minus .07 plus or minus .02g or greater the roll erection shall be reduced to cutoff. For lateral accelerations of .35 ± .05g, pitch erection shall be cut off.

3.5.1.1.3.4 Fore-Aft Acceleration - An accelerometer mounted on the Displacement Gyroscope with associated circuitry shall be provided to cut off the pitch erection during horizontal fore-aft acceleration. For aircraft longitudinal accelerations of plus or minus .07 plus or minus .02g or greater, the pitch erection shall be cut off.

3.5.1.1.3.5 Earth's Rate Compensation - Precision torques from latitude setting of Controller shall be applied to the vertical gyro to compensate for apparent drift due to the rotation of the earth.

3.5.1.1.4 Erection Cycle - Second Order - pitch and roll erection shall automatically sequence to the Second Order mode after the completion of the initialization sequence or completion of the fast erection cycle and when the system receives a valid velocity reference input. In this mode the reference velocities resolved by the system into a compatible coordinate frame shall be compared to the system velocities derived within the Second Order erection loops. Differences from the velocity comparisons shall be utilized to damp the Second Order erection control loop and produce the system vertical performance of 3.4.2.

3.5.1.1.5 Reference Velocity Preference - Upon completion of the First Order erection cycle (3.5.1.1), the system shall immediately switch to the damped Second Order using the valid velocity reference. If at the completion of First Order erection, the velocity reference source is unreliable, the system shall remain in the compensated First Order erection. Thereafter, if the reference velocity source is determined valid after 2 plus or minus 2 seconds the system shall automatically sequence to damped Second Order erection utilizing the valid reference velocity.

3.5.1.1.6 Roll Performance - The roll outputs shall provide a measure of angular displacement about the longitudinal axis of the aircraft. The assembly's roll-output angle shall increase for a rotation about the aircraft longitudinal axis that causes the right wing to drop below the horizontal plane. Roll shall have a range of plus or minus 180 degrees.

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3.5.1.1.6.1 Roll Accuracy and Threshold - The roll output shall be 0 plus or minus .20 degree when the assembly is mounted to a horizontal surface. The angular movement of plus or minus .05 degree in the roll axis about the roll level position shall be detectable in the roll outputs.

3.5.1.1.6.2 Roll Scale Error - The roll scale error shall not exceed 0 plus or minus .2 degree at a 0 degree roll angle and shall not exceed plus or minus .50 degree at roll angles in excess of plus or minus 5 degrees.

3.5.1.1.6.3 Roll Drift - The maximum drift in roll, exclusive of earth's rate, shall not exceed 6.0 degrees per hour. The maximum apparent free drift in roll shall not exceed 6.5 degrees per hour.

3.5.1.1.6.4 Roll Follow-Up Rate - The system shall be capable of continuously following a roll input rate of 600 degrees per second.

3.5. 1.1.7 Pitch Performance - The pitch outputs shall provide a measure of angle between the aircraft longitudinal axis and the local horizontal plane. The pitch output angle shall increase for an aircraft movement about the lateral axis that causes the nose of the aircraft to rise above the horizontal plane. The pitch output shall have a range of plus or minus 90 degrees.

3.5. 1.1.7.1 Pitch Accuracy and Threshold - The pitch outputs shall be 0 plus or minus .20 degree when the gyroscope assembly is mounted to a horizontal surface. The angular movement of 0 plus or minus .05 degree in the pitch axis about the pitch level position shall be detectable in the pitch outputs.

3.5.1.1.7.2 Pitch Scale Error - The pitch scale error shall not exceed 0 plus or minus .2 degree at a 0 degree pitch angle and shall not exceed plus or minus .5 degree at pitch attitude beyond 5 degrees.

3.5.1.1.7.3 Fitch Drift - The maximum free drift in pitch, exclusive of earth's rate, shall not exceed 6.0 degrees per hour. The maximum apparent free drift in pitch shall not exceed 6.5 degrees per hour.

3.5.1.1.7.4 Pitch Follow-Up Rate - The system shall be capable of continuously following a pitch input rate of 50 degrees per second,

3.5.1. 1.8 Azimuth Performance - The heading outputs shall provide a measure of angular displacement about the local vertical axis. The heading output angle shall increase for a clockwise movement of the aircraft about the vertical axis. Heading shall have a range of 360 degrees continuous.

3.5.1. 1.9 Initialization Cycle - Initialization of the directional gyro and the heading outputs shall commence automatically following application of power. Within the time specified in Paragraph 3.3.11, the system shall be operating in the mode selected on the Controller, meeting the requirements of Paragraph 3.4.1.

3.5.1.2 DG Mode Performance -

3,5.1.2.1 Heading Accuracy and Threshold - The scale error of the heading outputs shall be plus or minus .5 degree maximum. A movement of the Displacement Gyroscope assembly about the vertical axis equal to or greater than plus or minus .1 degree shall be detectable in the Heading outputs.

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3.5.1.2.2 Heading Drift - The maximum free drift in Heading with the system in the DG mode shall not exceed plus or minus 1.0 degree per hour where the earth's rate and drift bias are electrically compensated.

3.5.1.2.3 Drift Correction - The system shall contain earth rate and drift bias compensation for Directional Gyro drift. Drift bias shall have a range of plus or minus 4.5 degrees/hour and shall have an accuracy of plus or minus .2 degree/hour. Earth rate compensation shall be 15 degrees per hour times the sine of the latitude and shall have an accuracy of plus or minus .3 degree per hour.

3.5.1.2.4 Manual Slewing - Operation of the Set Heading control on the Controller shall allow slewing of the heading outputs at varying rates to 30 degrees per second. Settable accuracy shall be within plus or minus .1 degree.

3.5.1.2.5 Following-Up Rate. The heading outputs shall be capable of continuously following a heading input at a rate of 300 degrees per second.

3.5.1.3 SLAVED Mode Performance -

3.5.1.3.1 Heading Accuracy - The static and dynamic accuracy of the heading outputs after proper adjustment of the index, one cycle and two cycle compensation adjustments shall be as specified in 3.4.2.

3.5.1.3.2 Slaving Characteristics - The slaving rate of the heading loop shall be 1.5 degrees per minute per degree of synchronization error, up to a maximum slaving rate of plus or minus 2.5 plus or minus .2 degree per minute as specified in Figure 3. The synchronization error shall be filtered with a time constant of 5 plus or minus 1 second.

3.5.1.3.3 Slaving Cutoff

3.5.1.3.3.1 Turn Rate Cutoff - A solid state azimuth rate sensor shall be provided to permit control of heading slaving during turns. Under standard conditions, slaving cutoff shall occur for the following azimuth inputs:

a. Slaving shall be removed during turns of plus or minus 8 degrees, plus or minus 30 degrees, and plus or minus 180 degrees per minute within 30, 11, and 8 seconds respectively, and shall be returned within 20 seconds after the rates have ceased. Turn rate cutoff shall not be commanded by turns of plus or minus 3 degrees per minute or less.

b. Slaving shall be removed within 30 seconds when an input of 30 degrees per minute rate of turn is superimposed on a plus or minus .8 degree, one cycle per second azimuth oscillation. Slaving shall remain cutoff until the 30 degree per minute rate of turn has ceased.

c. Turn rate cutoff shall not be commanded by the following azimuth oscillations:

(i) plus or minus .8 degree with a 1.0 second period.

(ii) plus or minus 1.5 degrees with an 8.0 second period.

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3.5.1.3.3.2 Acceleration Cutoff - Heading slaving shall be cut off if either the longitudinal or the lateral acceleration exceeds .07g plus or minus .02g.

3.5.1.3.3.3 Attitude Extremes - Heading slaving shall be cut off at any time when either the roll or pitch attitude of the aircraft is not within plus or minus 25 plus 3 minus 0 degrees of horizontal.

3.5.1.3.4 Coriolis Compensation - The magnetic heading signal shall be compensated for errors due to Coriorlis acceleration using the reference velocity. Should the velocity reference input be invalid or not available, a fixed 300 knots velocity input shall be provided.

3.5.1.3.5 Fast Synchronization - Momentary depression of the Set Heading control on the Cotroller while in the Slaved Mode shall Provide automatic synchronization of the heading outputs to the magnetic heading at a slaving rate of not less than plus or minus 60 degrees per second. Fast slaving shall be automatically terminated when synchronization is complete to within plus or minus .25 degree. Magnetic fast synchronization shall be inoperative in the DG and Emergency modes or during slaving cutoff due to acceleration or attitude extremes.

3.5.1.4 EMERGENCY Mode Performance - The system shall provide electronically damped magnetic data when the Displacement Gyroscope is inoperative. The magnetic heading data shall be insensitive to normal magnetic field strength variations as specified herein.

3.5.1.4.1 Accuracy - When operated in the EMERGENCY mode, the heading outputs shall have an accuracy of 1.0 degree or less when referenced to the output of the compass transmitter in the standard magnetic field, and shall be damped electronically.

3.5.1.4.2 Backup Capability - In the EMERGENCY mode, the heading outputs shall operate normally for 95 percent or greater of any failures in the gyroscope assembly (whether BITE detected or not), and shall always operate normally if the cabling to the gyroscope assembly is disconnected.

3.5.1.5 Power Interruption

3.5.1.5.1 Erection Cycle - Means shall be provided to prevent the system from recycling through the initial fast erection sequence when power has been interrupted for a period of less than 60 seconds. For a power interruption of more than 90 seconds, the system shall automatically recycle through the fast erection sequence.

3.5.1.5.2 Attitude Information - After the initialization cycle is complete, any time the power is removed for 25 seconds or less, and the gyroscope displaced through angles of plus or minus 15 degrees in pitch and roll- and returned to its original vertical position, the pitch and roll axes shall indicate the original position to within plus or minus 1 degree within 10 seconds after reapplication of power.

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3.5.1.5.2.1 Heading Information - Means shall be provided for maintaining the heading synchronization through a complete power dropout of 60 seconds so that heading outputs after the power interruption is removed shall not differ from the heading outputs prior to the power interruption by more than 1.0 degree.

3.5.1.5.3 Damage Prevention - Means shall be provided to prevent damage to the gyroscopic assembly when it is blanked and pitched plus or minus 60 degrees or more during a power interruption of 0 to 90 seconds.

3.5.1.6 Malfunction Monitoring - Built-in-Test Equipment

3.5.1.6.1 Organizational Level BITE Provisions - The equipment shall include BITE provisions as defined herein per AR-10.

3.5.1.6.2 BITE - General - BITE provisions as defined herein shall accomplish the following:

(1) In-flight and organizational level fault detection

(2) In-flight and organizational level isolation of faulty weapons replaceable assembly.

3.5.1.6.3 BITE Methods - BITE shall make the maximum use of subsystem functional circuits to accomplish its purpose. The two forms of BITE are:

(1) Automatic continuous BITE (voltage measurements, etc) for all system functions practical.

(2) Ground, manually activated BITE shall be employed to monitor/test the detector.

The automatic continuous BITE shall not interfere with the normal Amplifier Control Assembly or remove the signals of the Compass Transmitter normal operation or cause incorrect data to be transmitted from the Electronic Control Amplifier to any interfacing equipment.

3.5.1.6.4 BITE Operating Modes - BITE operating modes shall be as follows:

(1) Automatic Operation - BITE shall operate continuously and automatically whenever the equipment is in operational status and shall provide the degree of fault detection capability specified herein.

(2) Manual Operation - The detector function shall be a manual mode test a ground operator activating the BITE switch.

3.5.1.6.5 BITE Performance -

3.5.1.6.5.1 Test Thoroughness - BITE provisions shall functionally test all system equipments. Testing of individual weapons replaceable units, subassemblies, circuit, etc, without assessing the overall functional capability of the equipments shall not be acceptable.

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3.5.1.6.5.2 Test Accuracy - The accuracy of BITE stimuli and measurements shall be adequate to preclude false pass/fail indications. BITE shall be designed to prevent fail indications due to primary power transients, switching transients, pass system operations, and other manual reference system application functions.

3.5.1.6.5.3 Fault Detection Capability - BITE shall detect and indicate 85 percent of failures of out-of-tolerance conditions in the equipments. These requirements apply to all equipments of the system, including 'compass Detector/ Transmitter failures, when BITE is required for in-flight and on-the-ground testing.

3.5.1.6.5.4 Fault-Isolation Capability - BITE shall isolate at least 85 percent of all detected faults to the faulty weapons replaceable assembly. This requirement applies when BITE is used for in-flight or on-the-ground fault-isolation capability and shall be provided to aid in isolation of intermittent faults. In-flight or on-the-ground isolation provisions shall be mechanized to prevent unsafe or ambiguous flight conditions. BITE shall not be responsible for fault isolation ambiguities resulting from open circuits, short. circuits, and ground in interconnecting harness external to the equipment. BITE shall not degrade the performance of any equipments of the system.

3.5.1.6.5.5 Automatically Monitored Functions - The following minimum functions shall be monitored for detection and isolation:

(1) Directional Gyro Spin Motor
(2) Vertical Gyro Spin Motor
(3) Outer Roll Servo Loop
(4) Pitch and Roll Torquers
(5) Accelerometers and Erection Amplifiers
(6) Roll Gimbal Servo Motor
(7) Vertical Gyro Gimbal Bearings
(8) Power Supplies
(9) Outer Roll Bearings
(10) Compass Transmitter or Detector Interface
(11) Vertical Gyro Erection Circuits
(12) Pitch, Roll, and Azimuth Output
(13) Initialization Circuits
(14) Mode Selection Switching
(15) Latitude Circuits
(16) Analog and Digital Circuits
(17) Pitch, Roll and Heading Resolvers, etc
(18) Heading Coupler
(19) Memory

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3.5.1.6.5.6 Manually Activated Monitored Functions - The following minimum functions shall be monitored for detection and isolation upon a manually activated command:

(1) Compass Transmitter or Detector

(a) Any failure in excitation coil
(b) Any short to ground
(c) Any open leg or lead

3.5.1.6.6 BITE Reliability

3.5.1.6.6.1 Failure - A failure shall be a condition that prevents the equipments from meeting the operational performance requirements. If certain failures or tolerance limits are to be considered noncritical and hence not necessary to be included in BITE tested features, modification of the population of failures used for BITE performance calculations shall be subject to the approval of the procuring activity when specified by the contract.

3.5.1.6.7 BITE Indication

3.5.1.6.7.1 Components - Failures within each component of this system shall be indicated when a failure is detected.

3.5.1.7 System Outputs

3.5.1.7.1 Synchro Type Format Data Repeaters - There shall be a solid state synchro type format repeaters as specified herein for each of the three axes. The synchro type data repeaters shall contain no moving parts and shall not require adjustment when transferred as interchangeable electronic assemblies.

3.5.1.7.1.1 Isolation - Each solid state synchro type format output, except for one of the heading outputs, shall be electrically isolated from all other system outputs.

3.5.1.7.1.2 Excitation - Attitude and heading outputs shall be excited by 115 volt, 400 Hz sources. The" excitation voltages shall conform to MIL-STD-704, Category B.

3.5.1.7.1.3 Gradient - Each solid state synchro type format output shall have a no-load, line-to-line voltage of 11.8 Plus or minus 1.4 volts (rms) with a phase angle of 9 plus or minus 5 degrees with respect to the excitation phase.

3.5.1.7.1.4 Load - The pitch, roll and primary heading synchro outputs shall be capable driving four control transformers (AY500-5 or equivalent). The two other isolated heaing outputs shall be capable of driving three control transformers (AY500-5 or equivalent). The non-isolated heading output shall be capable of supplying 250 milliampere rms per leg.

3.5.1.7.1.5 Short Circuit Protection - The non-isolated heading output shall be capable of withstanding a continuous short circuit to ground on one or both legs.

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3.5.1.7.2 Plate of Turn Function - The Electronic Control Amplifier shall incorporate the necessary electronics to provide derived rate of turn output to a turn indicator. The heading rate output shall be a direct current voltage source proportional to the heading rate of change. When loaded with 500 ohms, the voltage source shall provide 2 plus or minus .14 mA at 180 degrees per minute to the load. The output shall be accurate within 7 percent or .070 mA, whichever is larger from 0 to 225 degrees per minute. The output snail saturate for a rate of 400 degrees per minute.

3.5.2 Displacement Gyroscope Assembly - The Displacement Gyroscope assembly shall consist of hermetically sealed gyroscope subassemblies which in turn consist of one or two vertical gyroscopes and directional gyroscope elements mounted in a multiple gimbal set. The gimbal set shall incorporate a redundant roll gimbal to avoid the gimbal-lock position. The Displacement Gyroscope assembly shall be designed and constructed to meet the requirements of this specification with suitable accelerometers. The arrangement shall permit all-attitude operation and avoid any lock position at all times. No errors exceeding 1 degree and lasting more than 5 seconds shall be developed as a result of flight in the vertical climb and dive positions.The accelerometers shall provide the accuracy specified by Figure 1. Gimbal outputs shall be continuous through 360 degrees in azimuth, roll end pitch.

3.5.2.1 Function - The Displacement Gyroscope assembly provides a stabilized vertical reference for pitch and roll, and a stabilized horizontal reference for all headings in DG and SLAVED modes.

3.5.2.2 Weight - The weight of the Displacement Gyroscope assembly shall not exceed 13.5 pounds.

3.5.2.3 Form Factor - The Displacement Gyroscope assembly shall have a volume not exceeding 345 cubic inches. The external surfaces of this assembly shall be painted color 17043, in accordance with FED-STD-595. The Displacement Gyro form factor shall be in accordance with Figure 8.

3.5.2.4 Contents - The Displacement Gyroscope Assembly shall include assemblies for the following functions:

3.5.2.4.1 Vertical Gyro - The vertical gyros shall be suspended in the first vertical gyro gimbal. When so aligned, the spin axis of the vertical gyroscopic element shall be aligned with the local gravitational vector. It shall have sufficient angular momentum and shall be designed to meet the performance specified herein.

3.5.2.4.2 Directional Gyroscope - The directional gyroscope shall have sufficient angular momentum and shall be designed to meet the performance specified herein. The directional gyroscope shall be suspended in a gimbal system and stabilized by the inertia of the gyro action.

3.5.2.4.3 Vertical Gyro Erection - The vertical gyro erection mechanism shall incorporate separate systems for pitch and roll. Erection shall be accomplished by acceleration sensors and erection torquers. The erection sensors shall be positioned and aligned on the gimbals in a manner that will isolate their outputs from the pitch and roll displacements of the aircraft; in part particular, the output of the roll erection sensor shall not be affected by

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aircraft pitch attitudes. The erection system shall provide both initial, normal, fast and final erection circuitry.

3.5.2.4.4 Acceleration Sensors - Acceleration sensors shall provide a measure of lateral and longitudinal acceleration as specified in Figure 1.

3.5.2.4.5 Erection Torquers - Torque motors shall be affixed to the vertical gimbal set to provide erection precession control for the vertical gyro. These torque motors shall be capable of erection at normal or fast rates as specified herein.

3.5.2.4.6 Output Devices - Standard output devices shall be affixed to the gimbals to provide roll, pitch, and heading displacement outputs from the gyroscope assembly.

3.5.2.4.6.1 Roll Devices - The roll devices shall provide sine and cosine outputs of the measured roll angle. The roll devices shall be affixed to the first vertical gyro axis of freedom.

3.5.2.4.6.2 Pitch Devices - the pitch devices shall provide sine and cosine outputs of the measured pitch angle. The pitch devices shall be affixed to the second vertical gyro axis of freedom.

The heading output devices shall provide the sine and cosine output of the resolved heading. The devices shall be affixed to the pivot axis of the directional gyro gimbal so that the device shaft will provide a measure of angular displacement. Input to the heading devices shall be provided by the Electronic Control Amplifier.

3.5.2.4.7 Time Totalizing Meter - The Displacement Gyroscope assembly shall contain a time totalizing meter conforming to MS27651. The time totalizing meter shall be located on and shall be visible without disassembly of the gyroscope assembly case.

3.5.2.4.8 BIT Indicator - A BIT failure indicator shall be located on the Gyroscope Assembly case and shall be visible without disassembly of the case. Upon indication of a Gyroscope Assembly failure from the Electronic Control Amplifier, the indicator shall be tripped to a latched position. Less of electrical power to the Displacement Gyroscope and the slewing of directional heading and other system operational functions shall not be indicated as a failure.

3.5.2.4.9 DG Drift Bias Adjustment - A DG drift bias adjustment if required shall be provided on the Displacement Gyroscope to compensate for DG drift. The DG drift bias element shall be accessible from the outside of the sealed case. The element shall incorporate a removable cover to prevent inadvertent change in adjustment. The DG bias adjustment, as necessary shall provide a range of plus or minus 4.5 degrees per hour when interfaced with the Electronic Control Amplifier.

3.5.2.5 Case - The Displacement Gyroscope assembly case shall form an integral unit for mounting directly to the aircraft. The gyroscope assembly shall be provided with quick disconnect mounts that require no special tools for

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mounting or dismounting. The gyroscopic case shall be hermetically sealed. A combination of hermetical sealing techniques shall be utilized to meet the requirements of 3.5.2.5.2.

3.5.2.5.1 Filling Medium - The filling medium shall be at least 98 percent pure, free from dust particles, and shall contain not more than .006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be 95 percent helium and 5 percent carbon dioxide.

3.5.2.5.2 Sealing - The hermetic seal shall not be dependent upon materials that will be affected by any conditions to which the unit may be subjected. The leak rate shall not exceed 10-5 std cc/second at an absolute pressure of 12 to 14.7 pounds per square inch.

3.5.2.6 Mounting - The mounting of the Displacement Gyroscope assembly shall be as specified herein and shall be such that the displacement reference gyro shall not deflect by more than plus or minus .1 degree in roll, pitch, or azimuth as a result of accelerations or any vibrations between O and 18 Hz with a double amplitude of .020 inch. The torsional resonant frequency shall be greater than 18 Hz. The mounting shall be of the "quick disconnect or removal" design in acordance with Figure 12.

3.5.3 Electronic Control Amplifier - The Electronic Control Amplifier shall be of interchangeable plug-in type modules with maximum efficiency in the use of subassemblies.

3.5.3.1 Function - The Electronic Control Amplifier shall perform all functions necessary to interface with the Displacement Gyroscope, Compass Transmitter or Detector, and SOARS Controller. The Electronic Control Amplifier shall provide Ditch, roll, heading, turn rate and BITE.

3.5.3.2 Weight - The weight of the Elctronic Control Amplifier shall not exceed 14.5 pounds.

3.5.3.3 Form Factor - The Electronic Control Amplifier assembly shall have a volume not exceeding 400 cubic inches. The form factor shall conform to Figure 9.

3.5.3.4 Contents - The Electronic Control Amplifier shall include the assemblies required to provide the following functions:

1. Solid state electronic heading couplers
2. Displacement Gyroscope initialization controls
3. Outer Roll servo controls
4. Pitch and Roll erection controls
5. Compass Transmitter or Detector excitation and signal processing
6. SOARS Compensator interface
7. Annunciator drive to the Controller
8. Amplifier data processor
9. Turn Rate output

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100 Pitch, Roll and Heading outputs

110 Monitoring and BITE

120 System Power Supplies

13. System Interface circuits

14. Switching and Test Logic

3.5.3.4.1 Heading Coupler - The Electronic Control Amplifier shall provide a heading synchronizing angle to the Displacement Gyroscope assembly. This angle shall be modified as required by the selected heading modes. The drive capability of the signal shall be sufficient to transmit to the Displacement Gyroscope assembly.

3.5.3.4.1.1 Heading Coupler Action in DG Mode - When the DG mode is sellected on the system Controller, the synchronization angle will be modified:

(a) as commanded by the SET HEADING control on the SOARS Controller

(b) to correct for the apparent heading drift due to the earth's rotation at the latitude set on the Controller by reference to a voltage and polarity transmitter from the Controller

(c) to correct for mean directional gyro drift by reference to a voltage and polarity transmitted from the Displacement Gyroscope assembly.

3.5.3.4.1.2 Heading Coupler Action in SLAVED Mode - The heading coupler shall provide magnetic slaving action when the SLAVED mode is selected on the SOARS Controller.

3.5.3.4.1.2.1 Compass Transmitter or Detector Interface - The heading coupler shall interface with the Compass Transmitter or Detector to obtain a magnetic-heading-angle reference. The heading coupler shall provide filtering of the sine and cosine of magnetic heading prior to determining synchronization errors. The filtering break frequencies shall be selected to provide maximum reduction of Compass Transmitter oscillation while causing minimum magnetic heading damping. The magnetic heading shall be compensated for Coriolis acceleration. The interface with the Compass Transmitter shall allow use of conventional electrical compass-swinging equipment for compass calibration in aircraft.

3.5.3.4.1.2.2 Compass Transmitter or Magnetic Azimuth Detector Excitation - The Electronic Control amplifier shall supply 3.5 plus or minus .5 volt ac for the Transmitter/Detector at nominal aircraft input line voltage.

3.5.3.4.1.2.3 Magnetic Synchronization Errors - The error between system output heading and the magnetic reference shall be insensitive to the magnitude of the earth's magnetic field. This error shall be used to obtain synchronization between the system output heading and the reference magnetic-heading angle.

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3.5.3.4.1.2.4 Normal Synchronization - the synchronizing angle shall be incremented at the specified slaving rate in the direction that will reduce the synchronizing error to zero in a minimum period of time.

3.5.3.4.1.2.4.1 Synchronization Angle Modification - The synchronization angle shall be modified in the SLAVED mode to:

1. Correct for the apparent heading drift due to the earth's rotation as determined by the latitude set on the system Controller by reference to a voltage and polarity transmitted from the Controller.

2. Correct the mean directional gyro drift by reference to a voltage and polarity transmitted from the Displacement Gyroscope.

3. Provide slaving as specified herein.

3.5.3.4.1.2.4.2 Fast Synchronization - Circuitry shall be provided to cause synchronization of the azimuth output at a fast rate when the system is operating in the SLAVED mode and the SET HEADING switch on the controller is depressed.

3.5.3.4.1.3 Heading Coupler Action in EMERGENCY Mode - When the EMERGENCY mode is selected, the magnetic heading derived from the Transmitter/Detector shall be insensitive to the magnetic field strength and shall be transmitted directly to the heading synchro repeater electronics. The directional gyro output heading resolver shall not be used in the EMERGENCY mode. The index, one-cycle, and two-cycle error compensation shall remain in the magnetic heading loop.

3.5.3.4.1.4 Heading Coupler Action in SLAVED Mode - In the SLAVED mode the output heading shall be referenced to magnetic heading. True heading for internal functions shall be derived by combining the output heading and the magnetic variation.

3.5.3.4.2 Initialization Control - The Electronic Control Amplifier shall automatically sequence erection in pitch and roll following the application of power and upon a manual demand as specified herein. When the erection control senses a null at the time specified herein, normal erection shall begin automatically.

3.5.3.4.3 Outer Roll Servo Control - The Electronic Control Amplifier shall control the outer roll vertical gyro gimbal.

3.5.3.4.4 Electronic Repeaters - The solid state synchro type data repeaters in each of the three axes of the system shall contain no moving parts. The repeaters shall not require any adjustment when transferred as an electronic assembly.

3.5.3.4.5 Rate-of-Turn Function - The Electronic Control Amplifier shall incorporate the necessary electronics to provide derived rate of heading to the turn indicator.

3.5.3.4.6 Malfunction Monitor - BITE - The equipment shall include BITE provisions for organizational level maintenance. BITE shall obviate the need for any external test equipment for the performance of preflight, postflight, or

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inflight testing, alignment of equipment, exclusive of compass calibration, and shall provide fault isolation down to the defective weapons replaceable assembly.

3.5.3.4.6.1 BIT Indicator - A BIT failure indicator shall be provided on the Electronic Control Amplifier as specified herein and shall be visible without disassembly of the amplifier case. The indicator shall be tripped to a latched position upon command from the BIT monitor circuitry located in the Electronic Control Amplifier. Loss of electrical power to the Electronic Control Amplifier as specified herein shall not indicate a failure.

3.5.3.4.7 Mounting - Mounting for the Amplifier shall be of the "quick-disconnect or removal type in accordance with Figure 13.

3.5.3.4.8 Time Totalizer Meter - The Amplifier shall contain a Time Totalizer Meter conforming to MS27651 or MS17322.

3.5.3.4.9 Cooling Requirement - Cooling requirements shall be provided for the Electronic Control Amplifier as specified in 3.3.13.

3.5.3.4.10 Amplifier Data Processor -

3.5.3.4.10.1 Function - The amplifier data processor, contained within the SOARS Electronic Control Amplifier shall be a general purpose digital processor with sufficient capability to provide the system performance characteristics specified herein. The processor shall accept input parameters from the Gyro-scope, Compensator and Controller via appropriate circuits in the ECA. The processor shall perform the computations required by the heading slaving loop, the vertical erection loop, and built-in-test equipment.

3.5.3.4.10.2 Memory - The amplifier data processor shall contain at least 2,048 words of memory, with an expansion capacity of 50 percent. The memory shall be nonvolatile semiconductor. A data word shall be 16 bits in length. The memory cycle time shall be 2.0 microseconds or less. For each new contrac-tually assigned procurement activity aircraft application(s), a minimum of two pluggable module memories shall be made available for the development and resolution of each application's software requirements.

3.5.3.4.10.3 Arithmetic Unit - The amplifier data processor shall perform addition in hardware in less than 2.6 microseconds.

3.5.3.4.10.4 Program - The program shall provide for the amplifier data processor to perform all necessary calculations, arithmetic operations not hard-ware implemented, and logical decisions. The program shall also have the ability to determine velocity reference validity, independent of the invalid discrete as supplied.

3.5.3.4.10.5 External Analog Interface - Means shall be provided for data exchange from external analog references or sources of the internal digital format of the Electronic Control Amplifier Processor.

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3.5.4 SOARS Controller - The system Controller shall provide as a minimum the following required functions and indications for the operation of the SOARS.

1. Mode Selector switch
2. Fast Synchronization switch (pushbutton)
3. SET HEADING control
4. Synchronization Indicator
5. Latitude Manual control
6. Malfunction Warning lamp
7. Fast Erection Control
8. Magnetic Variation Control Switch
9. Hemisphere Selection Switch

3.5.4.1 Function Contents - The SOARS Controller allows the pilot to control the operation of the system. It contains a button to command fast erection in pitch and roll; a mode selection switch for DIvERGENCY, SLAVED, and DG modes of operation; a unit that will cause a fast synchronization in SLAVED mode, or can be used to set heading to any angle in DG mode; a latitude function to set in the proper earth's rate correction; an annunciator to indicate the error between output heading and magnetic heading in SLAVED mode; a magnetic variation function; and a system malfunction warning lamp.

3.5.4.2 Weight - The weight of the Controller shall not exceed 1.0 pound.

3.5.4.3 Form Factor - The SOARS Controller shall have a volume not exceeding 40 cubic inches. The Controller form factor shall be in accordance with Figure 10.

3.5.4.4 Controller Design - The SOARS Controller shall the designed in accordance with MIL-C-6781, Type I. Fasteners shall conform to MIL-F-25173 and plastic plate lighting shall conform to MIL-P-7788, Type IV except as follows:

1. The contrast, stray light, and dial visibility requirements of MIL-L-27160 shall apply. Stray light shall be measured 1 inch in front of the Controller panel.

2. The SOARS Controller lighting (red, 5.0 volts) shall meet the color requirements of MIL-L-25467C and the brightness requirements of MIL-P-7788.

3.5.4.5 Prime Standard - Panel lighting shall be compared to and shall balance in brightness and color with the prime standard. The prime standard shall meet the requirements of 3.5.4.4. Visual comparisons shall be made in a dark room at a distance of 2 ± 1/2 feet.

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3.5.4.6 Selector Switch Functions - A Mode Selector switch shall be provided for selecting any of the following modes of operation:

1. SLAVED: In the SLAVED mode, the system shall operate as a gyro stabilized magnetic compass system.

2. DG: In the free directional gyro mode the system shall provide gyro stabilized heading (manually set) and shall be operated independently of the remote compass transmitter. The directional gyroscope shall be corrected for earth rate effects as a function of latitude.

3. EMERGENCY: In the EMERGENCY mode the system shall be operated only as an undamped magnetic compass slaved to the Compass Transmitter. Use of this mode shall be limited to flight emergencies only and it shall not be used to accomplish aircraft mission requirements.

3.5.4.7 Synchronizer and Heading Set - Push-to-sync automatic control shall be provided to effect the fast synchronization and the heading set. When in the DG mode, the SET HEADING control markings shall indicate in which direction the control is to be turned to provide a decreasing or increasing heading. The mechanism shall, via operator control, provide a continuously variable drive rate in either direction. The drive rate shall be so damped as to permit accurate selection of the desired heading in a minimun period of time. The control shall be of the push-to-turn, spring-to-center type.

3.5.4.7.1 Push-to-Sync - When in the SLAVED mode and when the SET HEADING control is pushed, the appropriate push-to-sync circuits in the Electronic Control Amplifier shall be activated.

3.5.4.8 Latitude Correction Control - A manually set control shall be provided to correct the system for the apparent drift of the gyro due to the earth's rotation. The latitude selected shall be indicated by a presentation graduated from O to 90 degrees with a major division for each 10 degrees and a minor division for each two degrees. The correction shall be applied in both DG end SLAVED modes of operation, and in First or Second Order operation.

3.5.4.8.1 North and South Hemisphere Selection - The selection of the Northern or Southern Hemisphere shall be made by a manual selector switch and the selection shall be indicated by the display of letters N or S on the Controller panel.

3.5.4.8.2 Latitude Readout - The design shall eliminate a parallax error when the latitude is read from an angle above or below the readout. The latitude correction shall be applied in both DG and SLAVED modes of operation.

3.5.4.9 Synchronizing Indicator - A synchronizing meter shall be provided to indicate synchronization to within plus or minus 1/2 degree between compass system output and the Compass Transmitter, or electrical equivalent, when in the SLAVED mode of operation. When the system is not synchronized, the synchrorlizing meter shall indicate in which direction the system must be corrected in order to avoid ambiguity. The synchronizing indicator lighting and markings shall afford ease of viewing under all operational conditions. the synchronizing indicator needle shall be capable of setting to a synchronized null within 3 seconds or less after the SET HEADING button is depressed when the compass system output and the detector are 180 degrees out of phase. The sync

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indicator shall be of a meter type and shall have indicated on the panel a "Cross" on the right hand side and a "Dot" on the left hand side.. The manual heading set control knob shall have a "Cross" above and to the right and a "Dot" above and to the left.

3.5.4.9.1 Synchronizing Indicator Sensitivity - When the Compass Transmitter is rotated to a heading 10 degrees greater than the compass system output, the sync indicator shall move full scale to the left. When the Compass Transmitter is rotated to a heading 10 degrees less than the compass system output, the synchronizing indicator shall move full scale to the right.

3.5.4.10 Malfunction Warning Lamp - A red malfunction lamp shall be provided that will operate when a system malfunction is detected.

3.5.4.11 Vertical Gyro Fast Erection Control - A momentary pushbutton shall be provided that when depressed will automatically result in the pitch and roll erection system reverting to the fast erection rate for only the time necessary to complete the erection cycle. The Amplifier Control Assembly monitor valid signal shall indicate invalid for only the time the Control Assembly is in the fast erection mode or when erection is inoperable.

3.5.4.12 BITE - The BITE function shall be continuous.

3.5.4.13 Magnetic Variation Input Control - A switch shall be provided to permit the manual input of Magnetic Variation. The Magnetic Variation control shall be a three-position switch labeled 15E, O, and 15W.

3.5.4.14 Markings - All markings shall be so durable as to withstand usage encountered in service. Unless otherwise specified, the style and proportions of numerals and letters shall conform to MS33558.

3.5.4.15 Color - The background of all dials, counters, and other external surfaces not otherwise specified shall be lusterless black, Color No. 37038 of FED-STD-595. Knobs shall be gray, Color No. 36231 of FED-STD-595 and all markings shall be white, Color No. 37875 of FED-STD-595.

3.5.5 Electronic Compensator Unit - The Electronic Compensator Unit shall be a modular unit which shall be mounted in the aircraft to permit easy removal. The unit shall be of minimum size and weight and shall not exceed .5 pound. The Electronic Compensator Unit must be completely interchangeable with the other three component WRAs of this specification and shall not require any adjustment or readjustment when one or more of the above mentioned system components are removed and/or replaced. A compensator cutoff switch shall be provided as required in 3.4.9.4.

3.5.5.1 Compass Compensator - The Compass Compensator shall meet the following requirements:

3.5.5.1.1 Compensation - Heading compensation shall be provided to compensate for index, one-cycle and two-cycle errors.

3.5.5.1.1.1 Index Error Adjustment and Range - A maximum of one adjustment shall be required to compensate for the system index error. This index correction range shall be plus or minus 2.0 to plus or minus 4.0 degrees. When one- and two-cycle adjustments are made, the index error adjustment shall not require readjustment.

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3.5.5.1.1.2 N-S and E-W (one-cycle) Adjustments - Two adjustments will be required to compensate for one-cycle error. The degree of compensation specified herein shall apply for interconnecting lead lengths between the Compass Transmitter and the Electronic Control Amplifier of up to 200 feet of No. 28 wire. The Compensator shall provide two adjustments with the means to compensate for errors in the cardinal heading signals as produced by the heading system, including the compass transmitter signals. One adjustment device shall be capable of compensating for the N-S transmitter signal error and the other adjustment device shall be capable of compensating for the E-W transmitter signal error. They shall be independently adjustable and shall be designed to have minimum adjustment interference. Compensation shall be provided with less than .1 degree error.

3.5.5.1.1.3 Adjustment Range - With the Compass Transmitter set on O degree heading, the N-S adjustment shall be capable of changing the output signal plus or minus 12 degrees to plus or minus 15 degrees in standard field. With the. Compass Transmitter setting on 90 degrees, the E-W heading shall be capable of changing the output signal plus or minus 12 degrees to plus or minus 15 degrees in standard field. A clockwise rotation of each adjustment as specified above shall cause an increase in the resultant heading signal.

3.5.5.1.1.4 N-S and E-W Adjustment Interference - A full-scale rotation of the E-W adjustment screw shall not influence the system heading output signal by more than plus or minus 1.5 degrees when the Compass Transmitter is set to a O degree heading. When the Compass Transmitter is set at a 90 degree heading, a full-scale rotation of the N-S adjustment screw shall not influence the system heading output by more than plus or minus 1.5 degrees.

3.5.5.1.1.5 Two-Cycle Adjustments and Range - A minimum of two adjustments shall be provided for two-cycle compensations. The cardinal and intercardinal adjustments shall have an adjustment range from plus or minus 2 degrees to plus or-minus 4 degrees.

3.5.5.2 Compensator Function - Means shall be provided to compensate for index, one cycle, and two cycle errors. The potentiometers shall be of the multi-turn type, 'and shall be mounted with the adjustment screws immediately behind the removable cover. The Compensator shall also include a fast swing switch, which, when activated, shall reduce the heading coupler time constant to ease compass calibration.

3.5.5.3 Weight - The weight of the Compass Compensator shall not exceed .5 pound.

3.5.5.4 Form Factor - The Compensator shall have a volume not exceeding 18 cubic inches. The form factor shall conform to Figure 11.

3.6 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (see Paragraph 6.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified, the supplier is responsible for the performance of all test requirements as specified herein. Except as otherwise specified, the supplier may use his own facilities or any commercial laboratory acceptable to the Government or the procuring activity. The Government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests (4.2)

(2) Initial Production Tests (4.3)

(3) Acceptance Tests (4.4)

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.5. The procurement activity representative shall be advised when tests are to be conducted so that a Government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Govenment or the procurement activity.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review in accordance with MIL-STD-333 and MIL-STD-1304.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all requirements of this specification, other applicable specifications and the contract. Preproduction tests shall include, but not necessarily be limited to electrical (bench), environmental, electromagnetic interference, reliability development, and maintainability demonstration tests.

4.2.2.1 Electrical (Bench) Tests - Bench tests shall be conducted to demonstrate compliance with specified performance parameters. These tests shall be conducted under standard laboratory conditions. Inspection and examination to determine compliance with material and workmanship requirements shall be conducted at this time.

4.2.2.2 Environmental Tests - Environmental tests shall be conducted in the sequence specified in MIL-T-5422. Environmental tests shall include the following:

4.2.2.2.1 Temperature/Altitude Test - A temperature/altitude test shall be performed in accordance with MIL-T-5422 for Class II equipment except that the short-duration high temperature operational test shall be conducted for one +95°C 30 minute period.

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4.2.2.2.2 Vibration Test - A vibration test shall be conducted in accordance with MIL-STD-810C, Method 514.2, Category C for equipment mounting _ configuration without vibration isolators.

4.2.2.2.3 Shock Test - A shock test shall be performed in accordance with MIL-T-5442.

4.2.2.2.4 Humidity Test - A humidity test shall be performed in accordance with MIL-T-5422.

4.2.2.2.5 Salt Fog Test - A salt fog test shall be performed in accordance with MIL-T-5422.

4.2.2.2.6 Explosion Test - An explosion test shall be performed in accordance with MIL-T-5422.

4.2.2.2.7 Acoustical Noise Test - An acoustical noise test shall be performance in accordance with MIL-STD-810 and Paragraph 3.3.10.1 of this specification.

4.2.2.2.8 Sand and Dust Test - A sand and dust test shall be conducted in accordance with MIL-T-5422.

4.2.2.2.9 Fungus Test - A fungus test shall be performed in accordance with MIL-T-5422.

4.2.2.2.10 Temperature Shock Test - A temperature shock test shall be performed in accordance with MIL-T-5422.

4.2.2.3 Electromagnetic Interference/Compatibility Test - An electromagnetic interference/compatibility test shall be performed in accordance with MIL-STD-461 and 3.3.6 according to measurement procedures defined by MIL-STD-462.

4.2.2.4 Reliability Development Test - Reliability development testing shall be performed In accordance with MIL-STD-2068(AS) utilizing a minimum of ten (10) equipments. Test duration shall be 20,000 hours. Test Level E per MIL-STD-781 shall be used except that the temperature extremes shall be +55°C and -44°C and the vibration .5g. The equipment utilized shall be refurbished by the manufacturer and the disposition shall be in accordance with the contract.

4.2.2.5 Maintainability Demonstration - Maintainability Demonstration shall be in accordance with MIL-STD-471 except as specified herein. Maintainability Demonstration shall be performed on a minimum of three selected pairs. The tasks to demonstrate maintainability shall be selected in advance of timing the tasks and shall be approved by the procuring activity.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. Any configuration changes resulting from the preproduction tests shall be incorporated into previously delivered equipment in accordance with the contractor support agreement with the procuring activity.

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4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government or the procurement activity, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - Requirements for initial production tests shall be satisfied by preproduction (First Article) tests as described in Paragraph 4.2.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the procurement activity or the Government representative. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results, sample and reliability assurance tests in a format acceptable to the procuring activity. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests	(4.4.1)
(2) Sampling Tests	(4.4.2)
(3) Reliability Assurance Tests	(4.4.3)
(4) Special Tests	(4.4.4)

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests.

(1) Examination of Product	(4.4.1.1)
(2) Operational Test	(4.4.1.2)

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check all characteristics and record all data to assure satisfactory equipment operation.

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4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests in accordance with the following:

Quantity of Equipment	Quantity to be Selected
First 51	1
Next 75	1
Next 100	1 for each additional 200 or fraction thereof

One out of first 51 need not be selected and tested if preproduction (First Article) tests are conducted.

4.4.2.1 Scope of Tests - Each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete all operational tests in accordance with Part I of the preproduction test procedure at ambient room conditions.

(2) Operational test during the temperature-altitude and vibration environments in accordance with Part II of the approved preproduction test procedures.

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and MIL-STD-2074(AS).

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled, "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. The Accept/Reject Criteria for Test Plan III shall be used.

4.4.3.2 Reliability Assurance Production Acceptance Phase - Reliability Assurance Production Acceptance Testing shall be conducted through a production burn-in. Each equipment produced shall be operated for 100 operating hours burn-in, and the test data (operating hours and failures) shall be totaled and compared to the Accept/Reject Criteria of Test Plan II of MIL-STD-781. (Extend the lines as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: AIR-53354B Avionics Division. Delivery of equipment shall not be held up to await a decision under the test plan. In the event an Accept Decision is reached, the test duration of each equipment shall be reduced to 50 operating hours. At any time that the current total of test hours and test failures plotted on Test Plan IV curves (extended) show a Reject Situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a Reject Situation exists pending a review of the

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contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc, so that the entire compilation will show other than a rejection decision. For the Production Acceptance Phase, Test Level E shall be used, except that the temperature extremes shall be +55°C and -44°C, and vibration shall be .5g. Details of the burn-in tests shall be included in the test procedures which are submitted and approved by the procuring activity.

4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc, shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) One an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of tests and test procedures previously approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure

(2) Determine if the failure is an isolated case or design defect.

(3) Submit proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests to the procuring activity for approval.

(4) Where practical, include a test within the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.5 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided by and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary. Interim test procedures for initial units shall be prepared to approved commercial standards. Test procedures per MIL-T-18303 shall be prepared under the procuring activity's support contract and upon approval shall immediately be utilized for acceptance of all subsequent units.

4.5.1 Internal Examination - All preproduction equipment shall be opened and the internal mechanism shall be examined to determine conformance with the applicable requirements of the specification. In addition, any damage or deterioration to the equipment resulting from any of the preceding tests, or any other deficiency which could in any manner prevent the instrument from meeting operational or maintenance requirements during service life shall be cause for rejection.

4.6 Reconditioning of Tested Equipment - Equipment that has been subjected to acceptance and special tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking the contractor shall resubmit the equipment for acceptance by the procuring activity. Equipment that has been subjected to preproduction and initial production tests shall be evaluated by the contractor and any recommendations regarding replacing worn or damaged items shall be submitted to the procuring activity for possible reconditioning under the applicable support contract.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements specified herein.

4.8 Rejection and Retest - Equipment that has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the Government representative.

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5. PREPARATION FOR DELIVERY

5.1 Packaging - All major units and parts of the equipment shall be packaged in accordance with MIL-E-17555 and MIL-STD-794, Level A. The method of preservation for Level A packaging shall be Method II of MIL-P-116. When specified by the contract, the displacement gyroscope assembly shall be packaged per MIL-G-0081559B(AS) except latches shall be used in lieu of screw fastenings.

5.2 Markings - All major components shall be marked as specified in MIL-STD-794 with an additional precautionary marking appearing conspicuously on two opposite sides of each Displacement Gyroscope and its shipping containers: FRAGILE - HANDLE WITH LOVING CARE - GYROSCOPIC NAVIGATION EQUIPMENT.

6. NOTES

6.1 Intended Use - This aircraft system provides velocity damped heading, pitch and roll, information to other flight, navigational and weapon delivery equipments and for direct display instrumentation to the pilots and crew.

6.2 Ordering Data - Procurement documents should specify the following:

(1) Title, number, and data of this specification.

(2) Selection of applicable levels of packaging and-packing shall be in accordance with Section 5.

(3) Data Requirements (See 3.6).

(4) Addition of mounting trays.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification- This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity.of design, performance, ease of maintenance and reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired that their use be investigated. When investigation shows advantages can be realized, a request

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for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Type Designations - The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 Revisions - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.7 Associated Equipment - The equipment shall operate properly with the following associated equipment that will not be supplied:

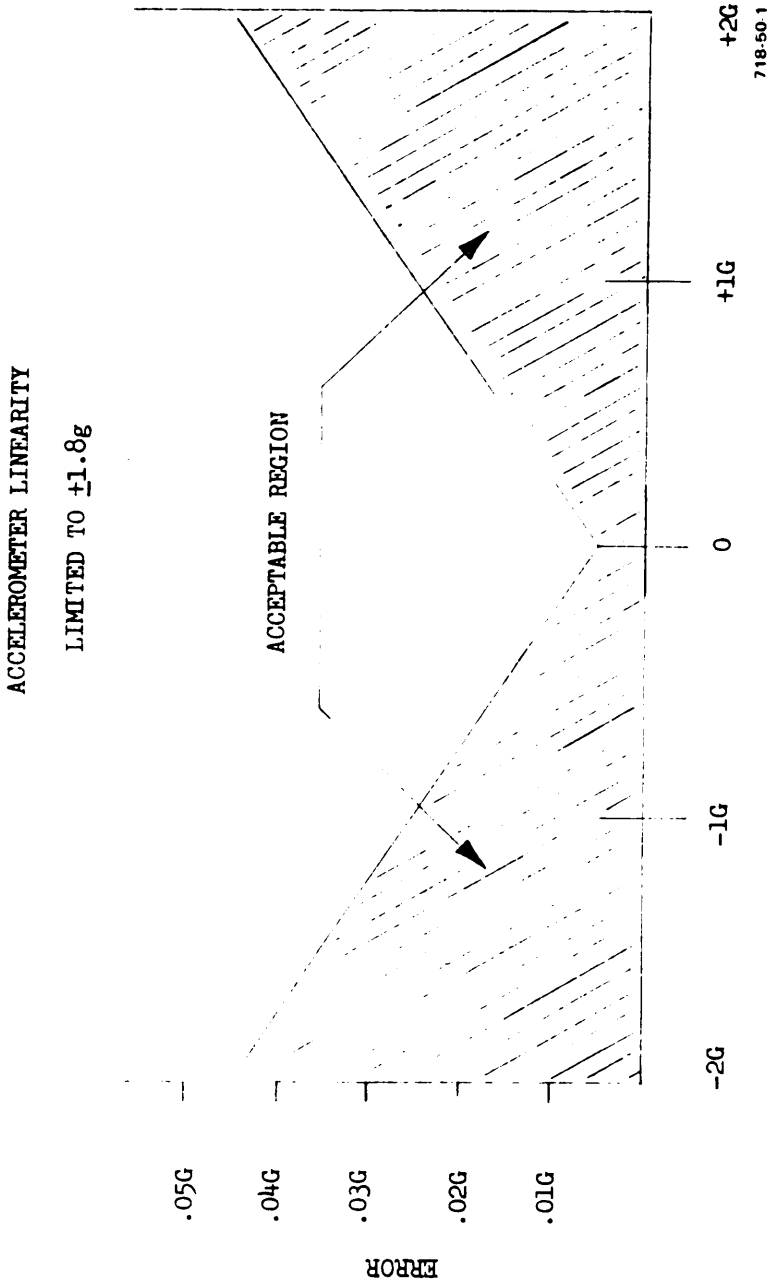
Item	Type Designation
Magnetic Compass Calibrators	MC-2
Magnetic Azimuth Detector	DSU-4A/A
Transmitter, Induction	TRU-79A/A
Compensator, Compass	MX-394A
Transmitter, Compass Remote	ML-1

6.8 This specification is under the cognizance of AIR-53354B.

Preparing Activity:
Navy- AS

Project No. 6605-N296

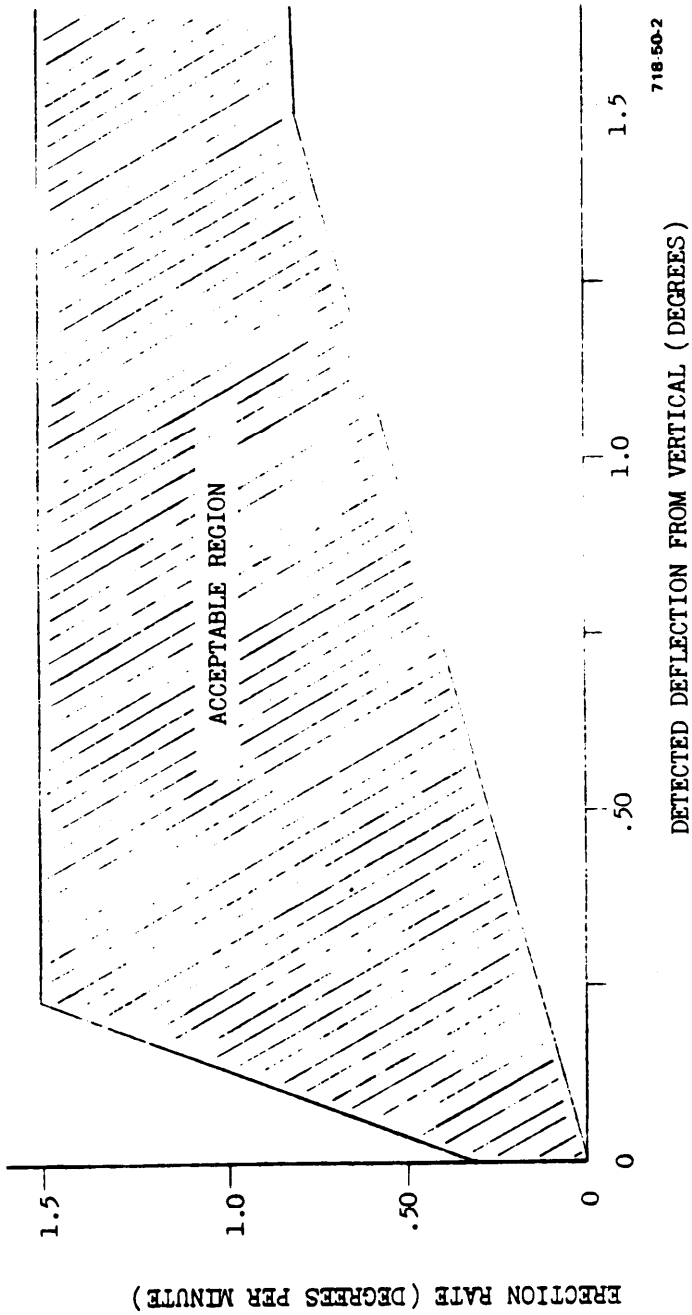
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MINIMUM ACCEPTABLE ACCELEROMETER CRITERIA

FIGURE 1

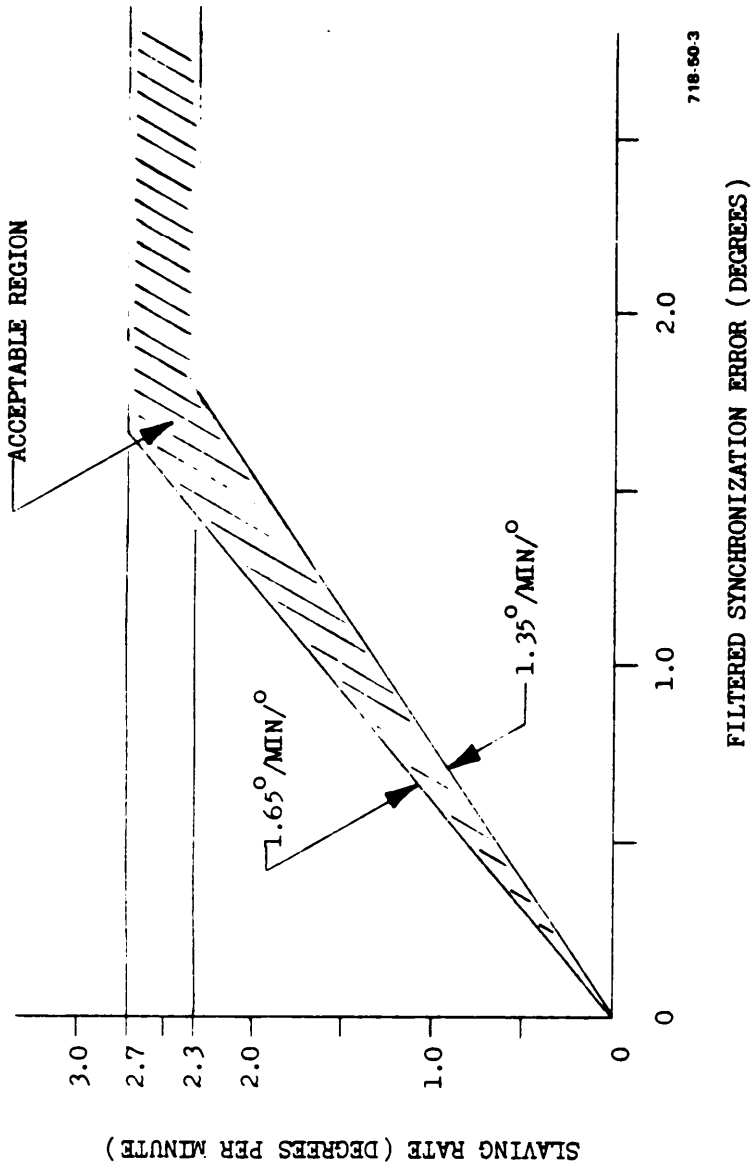
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NORMAL ERECTION RATE

FIGURE 2

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HEADING SLAVING RATE

FIGURE 3

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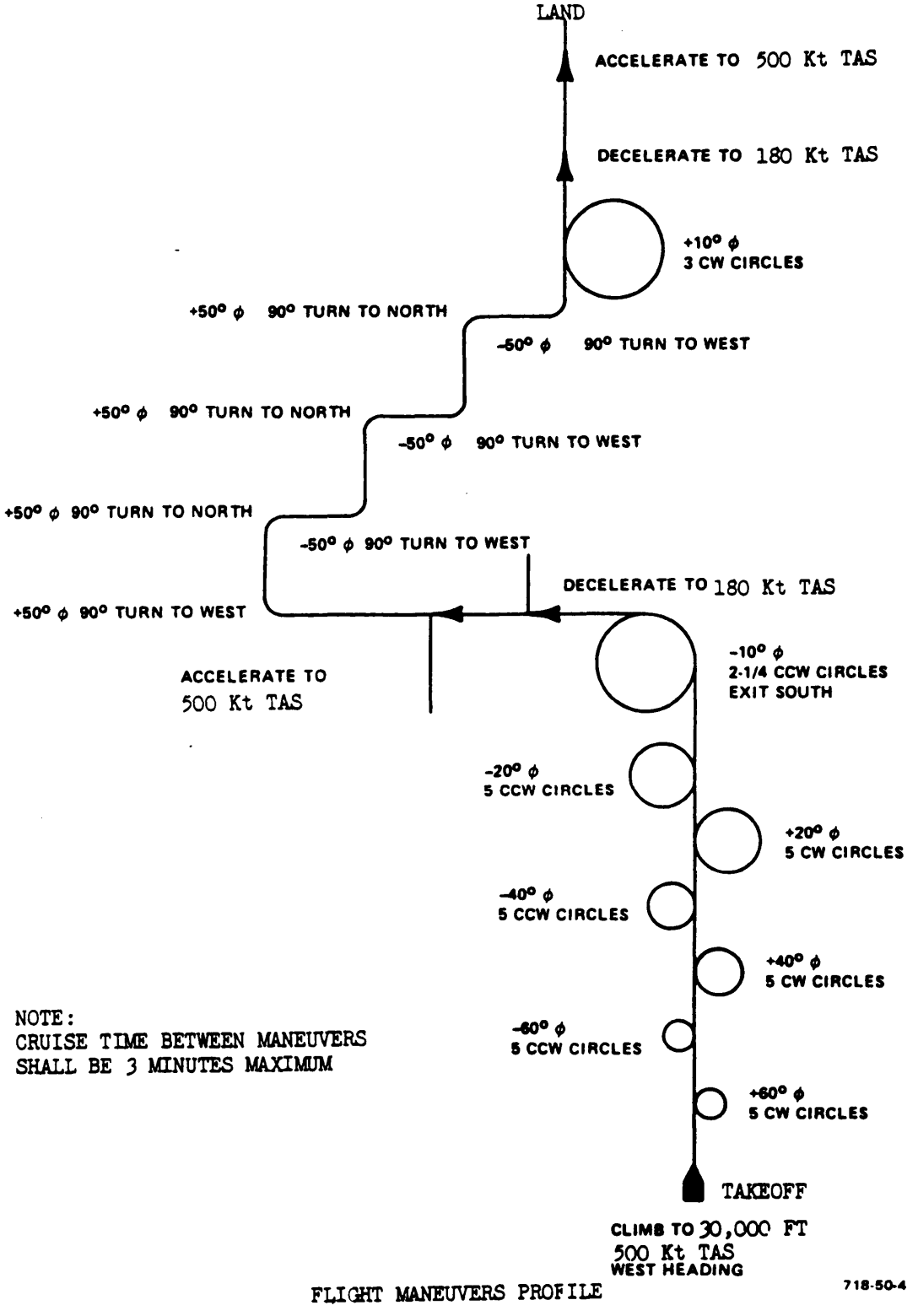
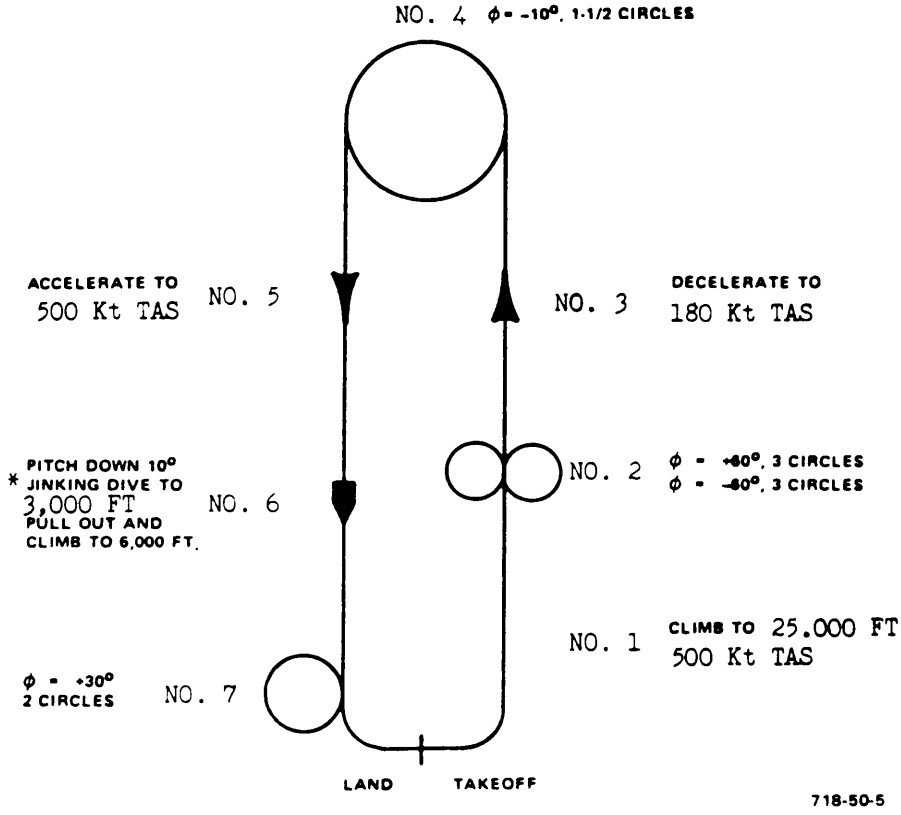


FIGURE 4

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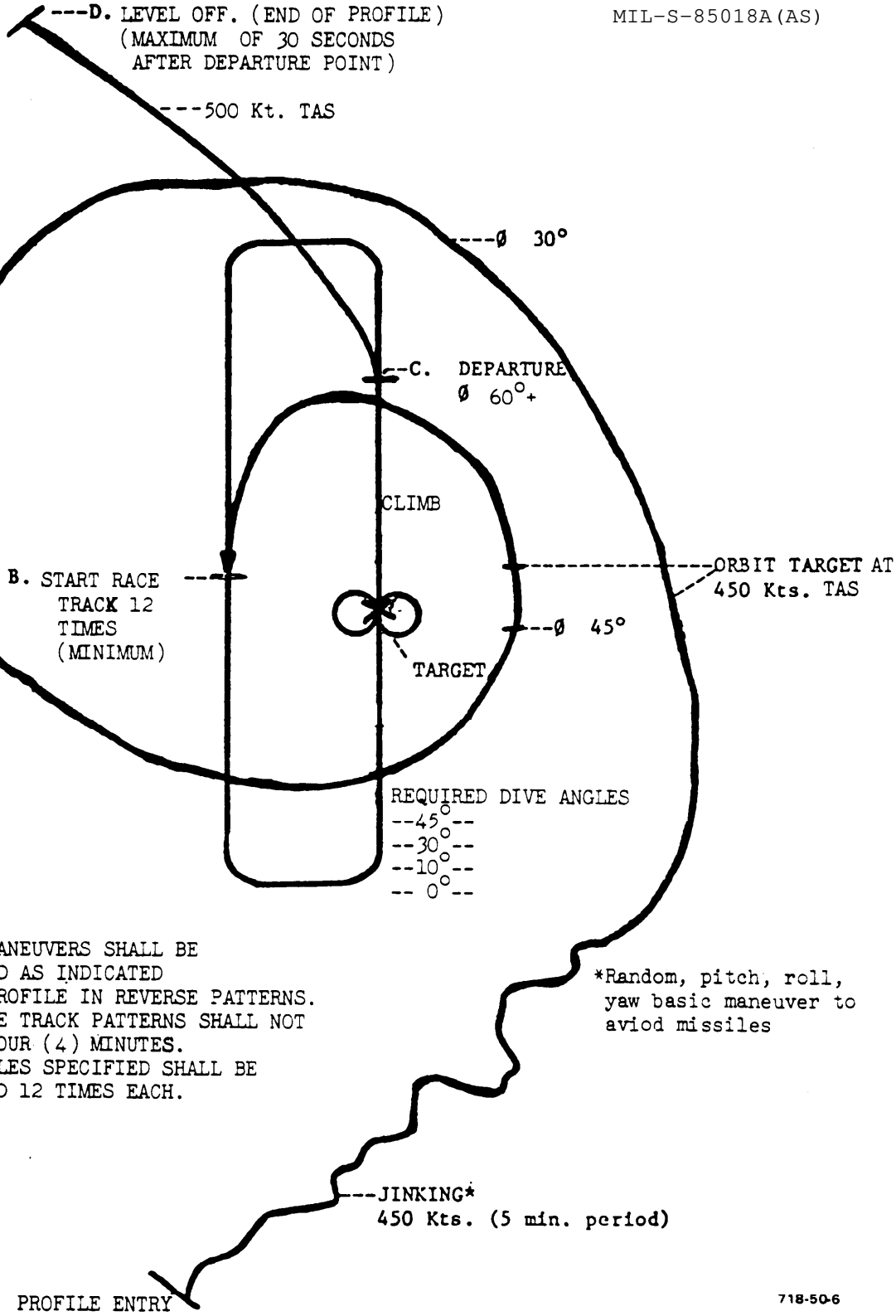
FLIGHT MANEUVER PROFILE

FIGURE 5

*Random pitch, roll and yaw evasive combat action

Note:
Cruise time between maneuvers shall be 3 minutes maximum.

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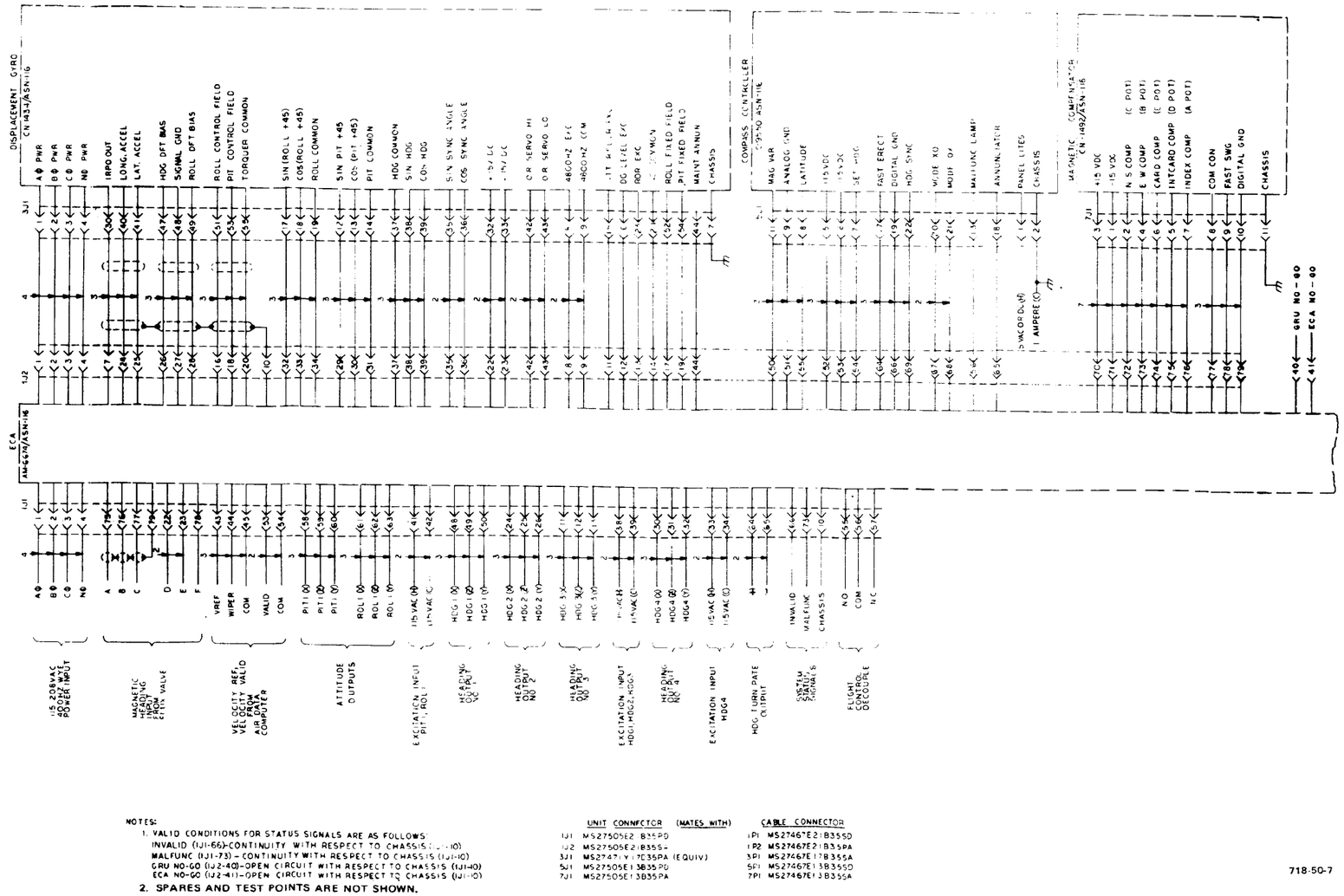
NOTE:
1. CRUISE MANEUVERS SHALL BE CONDUCTED AS INDICATED
2. REPEAT PROFILE IN REVERSE PATTERNS.
3. EACH RACE TRACK PATTERNS SHALL NOT EXCEED FOUR (4) MINUTES.
4.. DIVE ANGLES SPECIFIED SHALL BE CONDUCTED 12 TIMES EACH.

FLIGHT PROFILE

FIGURE 6

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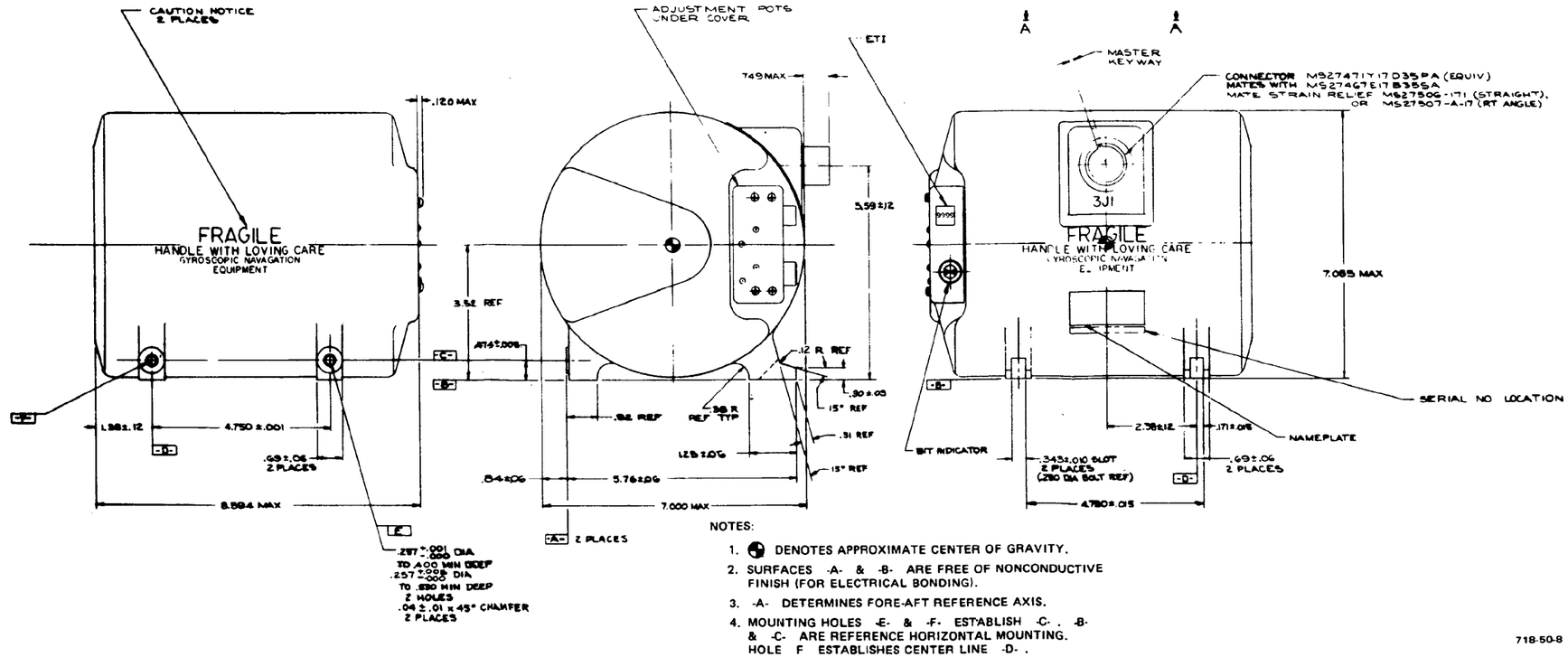
NOTES:
1. VALID CONDITIONS FOR STATUS SIGNALS ARE AS FOLLOWS:
   INVALID (1J1-66)-CONTINUITY WITH RESPECT TO CHASSIS (1J1-10)
   MALFUNC (1J1-73) – CONTINUITY WITH RESPECT TO CHASSIS (1J1-10)
   GRU NO-GO (1J2-40)-OPEN CIRCUIT WITH RESPECT TO CHASSIS (1J1-10)
   ECA NO-GO (1J2-41)-OPEN CIRCUIT WITH RESPECT TO CHASSIS (1J1-10)
2. SPARES AND TEST POINTS ARE NOT SHOWN.

	UNIT CONNECTOR	(MATES WITH)	CABLE CONNECTOR
1J1	MS27505E21B35PD		1P1 MS27467E21B35SD
1J2	MS27505E21B35S-		1P2 MS27467E21B35PA
3J1	MS2747[illegible]7C35PA	(EQUIV)	3P1 MS27467E17B35SA
5J1	MS27505E13B35PD		5P1 MS27467E13B35SD
7J1	MS27505E13B35PA		7P1 MS27467E13B35SA

718-50-7

FIGURE 7
SYSTEM INTERCONNECT WIRING

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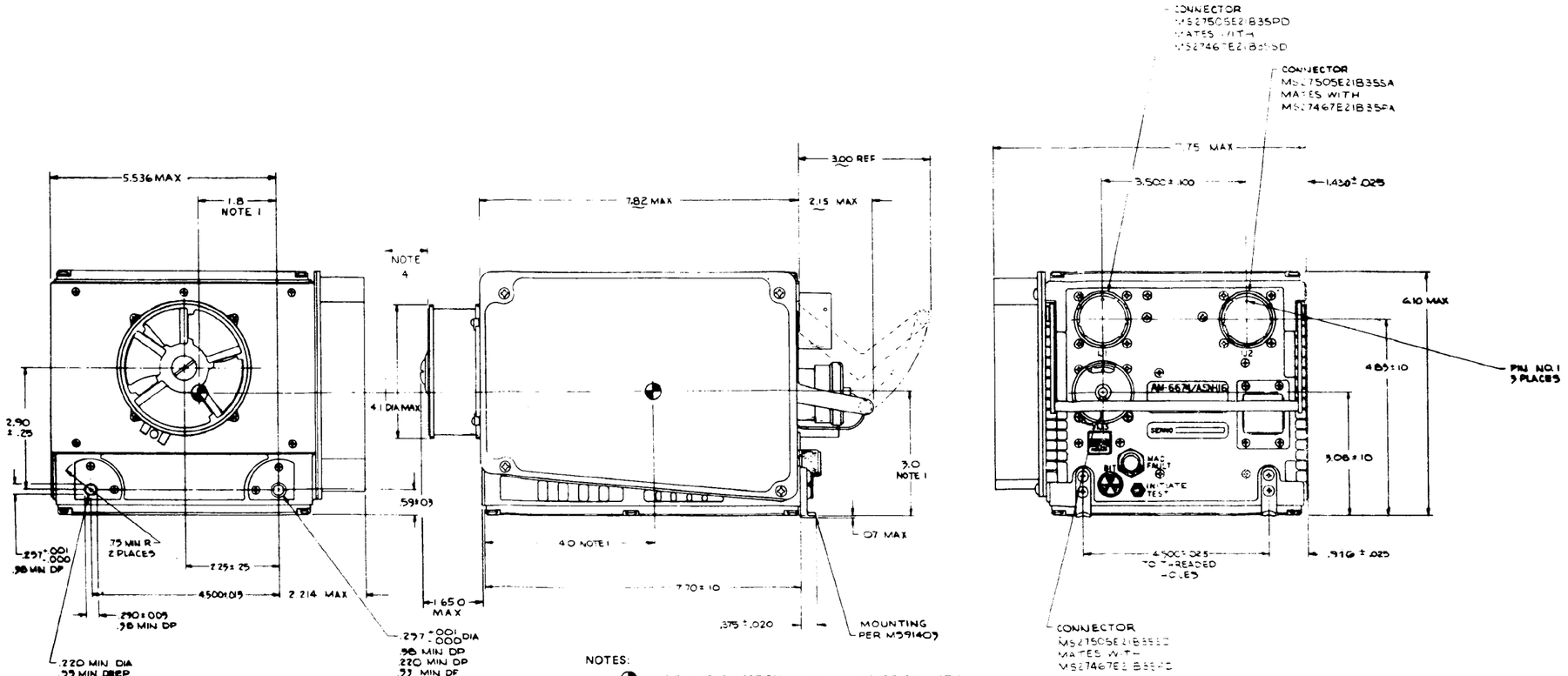
NOTES:

1. ⊕ DENOTES APPROXIMATE CENTER OF GRAVITY.
2. SURFACES -A- & -B- ARE FREE OF NONCONDUCTIVE FINISH (FOR ELECTRICAL BONDING).
3. -A- DETERMINES FORE-AFT REFERENCE AXIS.
4. MOUNTING HOLES -E- & -F- ESTABLISH -C- . -B- & -C- ARE REFERENCE HORIZONTAL MOUNTING. HOLE F ESTABLISHES CENTER LINE -D- .

718-50-8

FIGURE 8
DISPLACEMENT GYRO FORM FACTOR

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NOTES:

1. ◓ = INDICATES APPROXIMATE CENTER OF GRAVITY.
2. TWO INCH MINIMUM CLEARANCE REQUIRED FOR PROPER AIR FLOW.

718-50-9

FIGURE 9
ELECTRONIC CONTROL
AMPLIFIER FORM FACTOR

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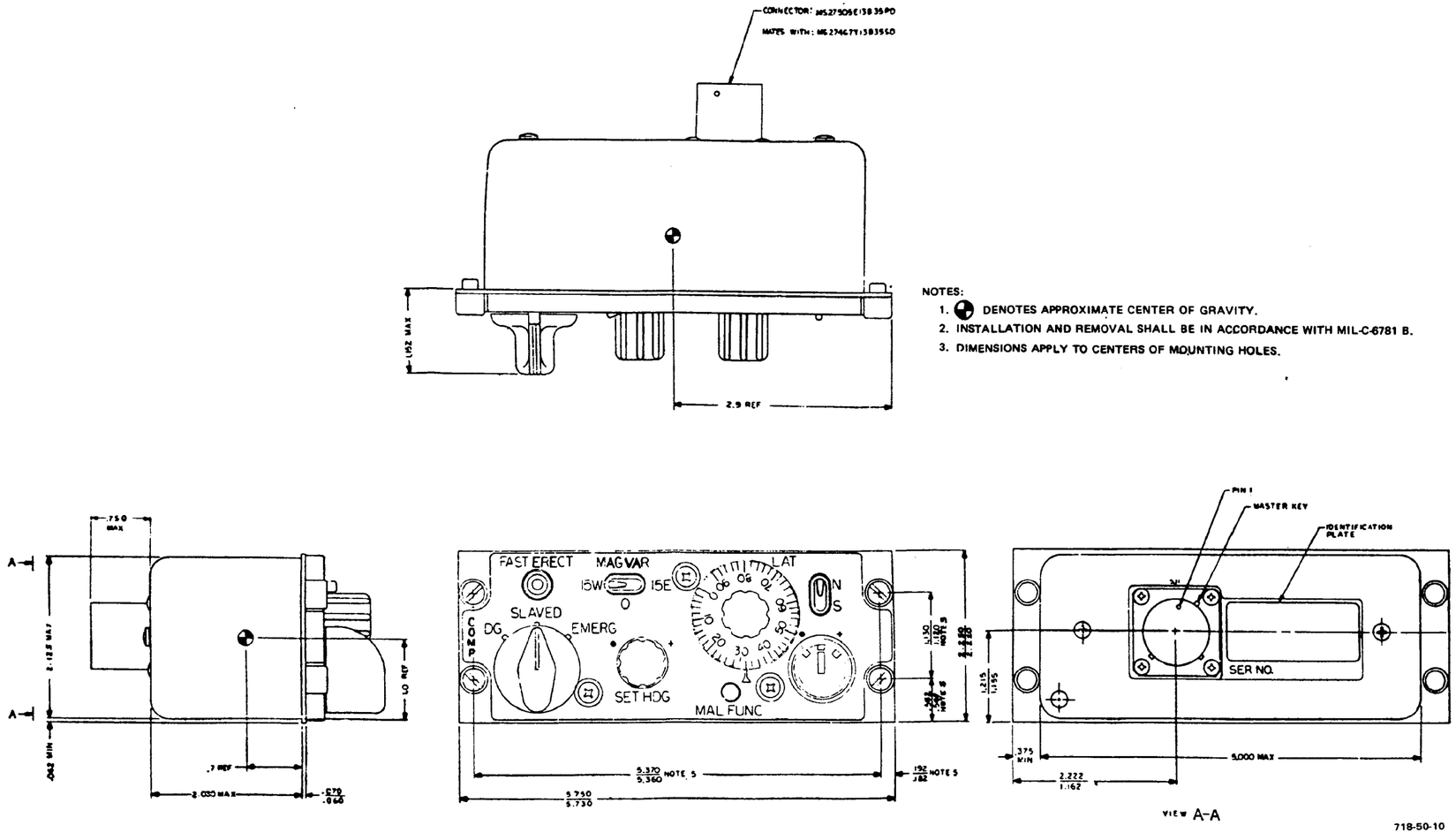


FIGURE 10
SOARS CONTROLLER FORM FACTOR

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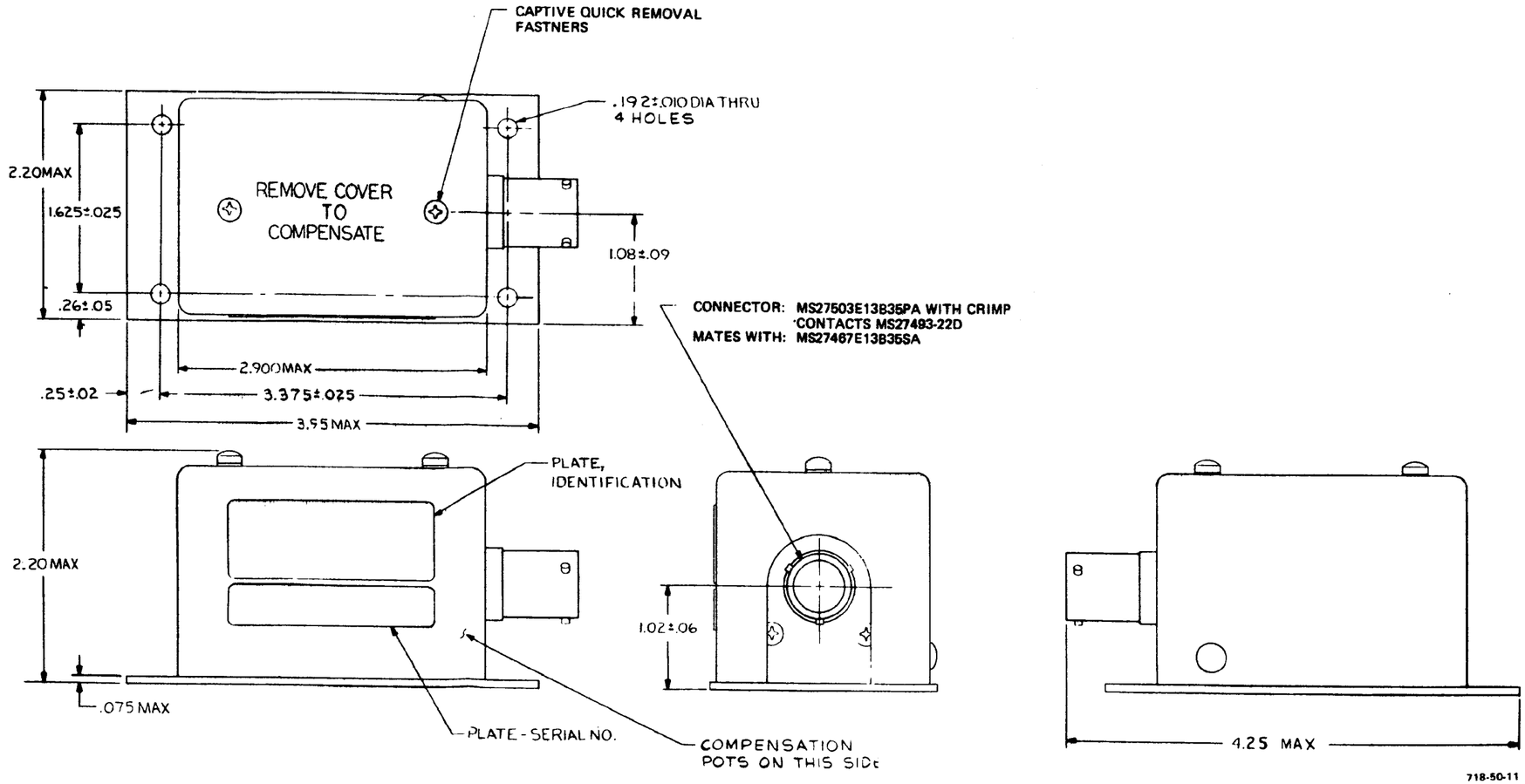
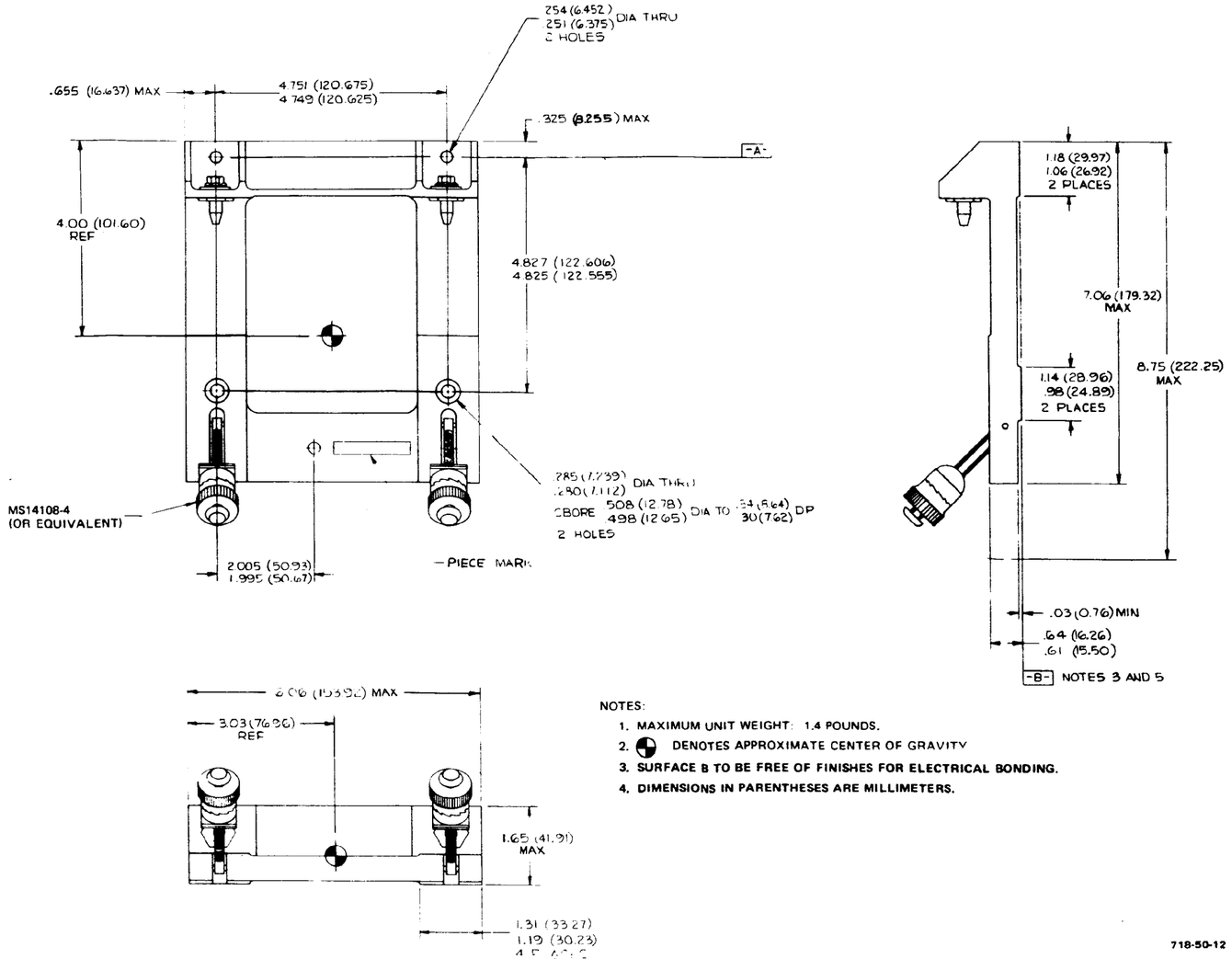


FIGURE 11
ELECTRONIC COMPENSATOR FORM FACTOR

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NOTES:

1. MAXIMUM UNIT WEIGHT: 1.4 POUNDS.
2. ⊕ DENOTES APPROXIMATE CENTER OF GRAVITY
3. SURFACE B TO BE FREE OF FINISHES FOR ELECTRICAL BONDING.
4. DIMENSIONS IN PARENTHESES ARE MILLIMETERS.

718-50-12

FIGURE 12
MOUNTING TRAY,
DISPLACEMENT GYRO ASSEMBLY

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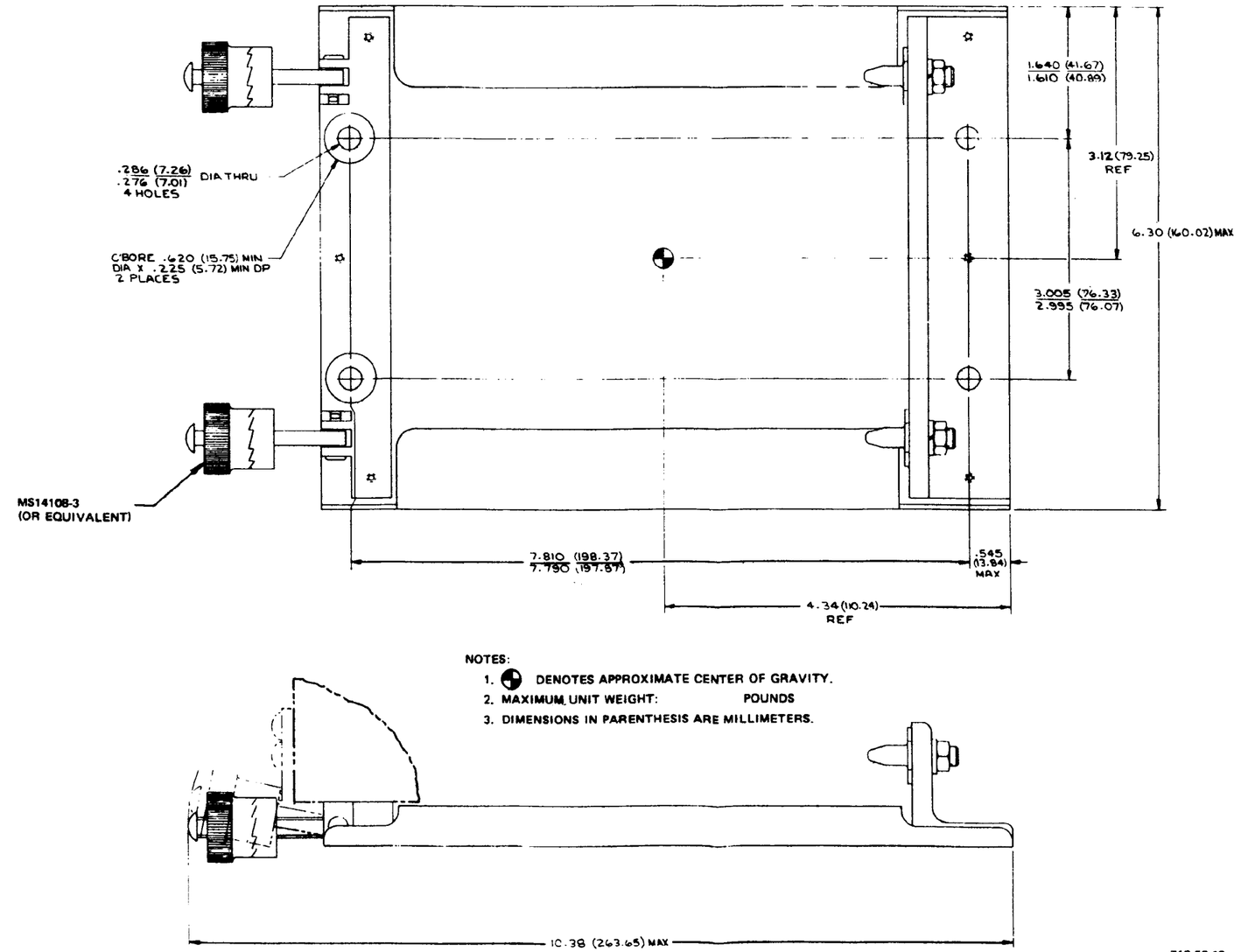


FIGURE 13
MOUNTING TRAY,
ELECTRONIC CONTROL AMPLIFIER

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CONTRACT NUMBER

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